



Fundamentals of neurobiology

ch4 : synapse & dendrite

R.C. Lambert

Neuronal networks and physiopathological rhythms

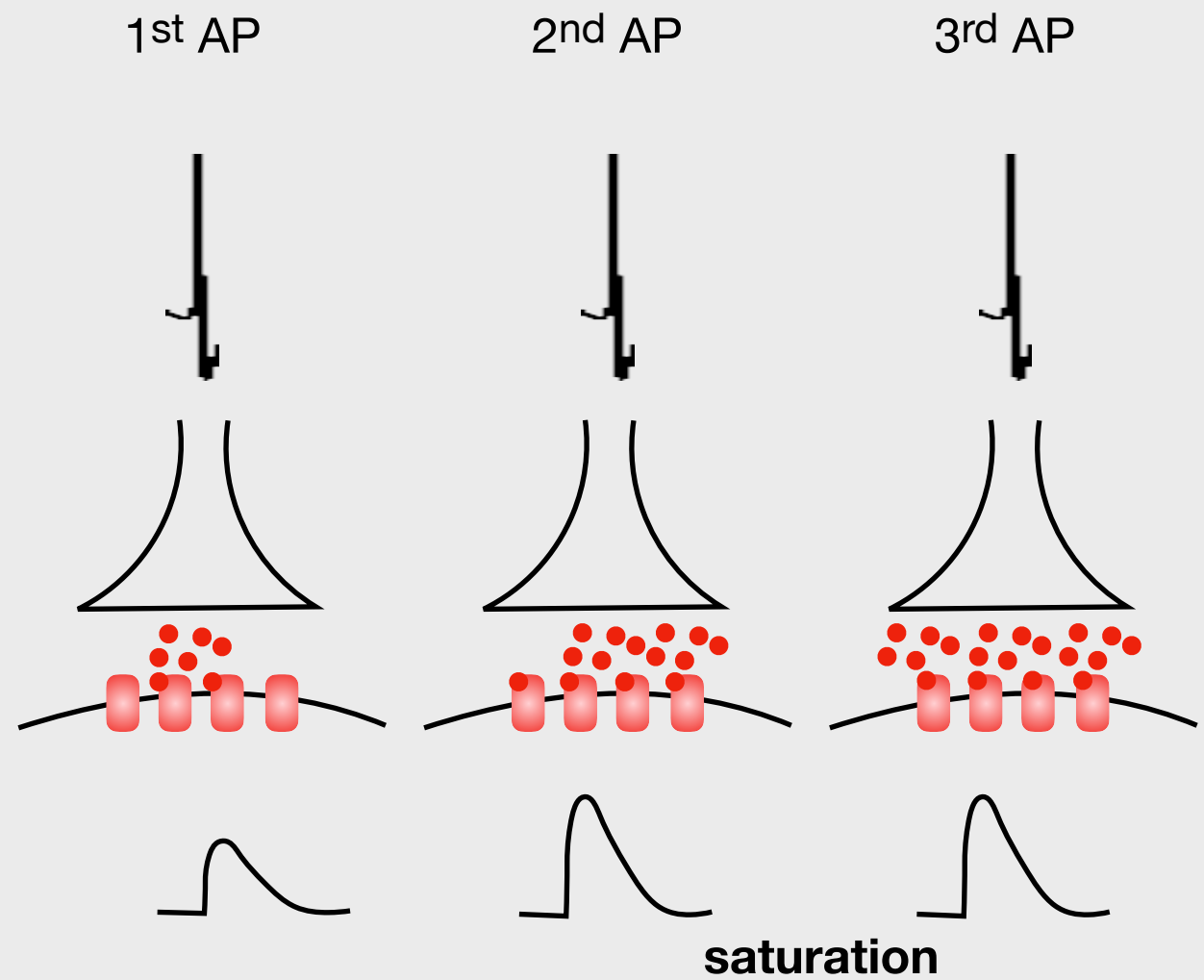
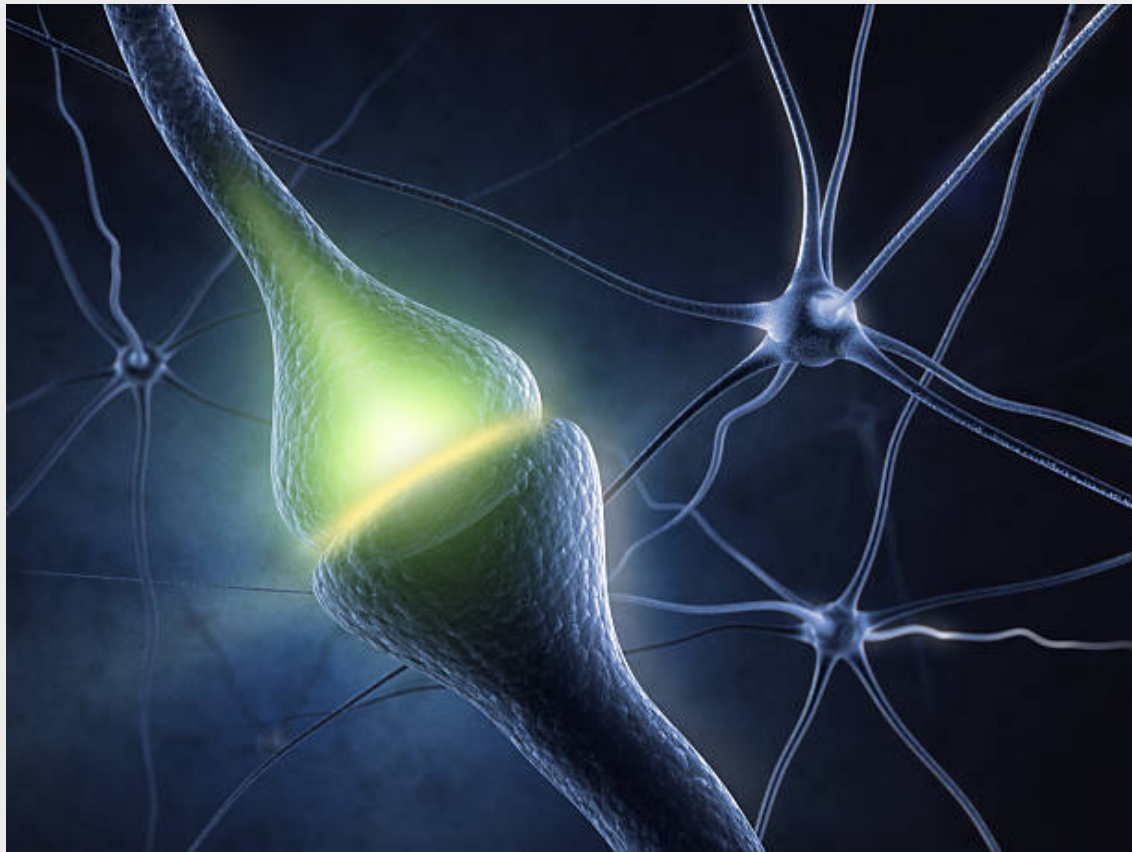
<http://www.ibps.upmc.fr/fr/Recherche/umr-8246/reseaux-de-neurones-et-rythmes-physiopathologiques>

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CNRS UMR8246, INSERM U1130, UPMC UMI19



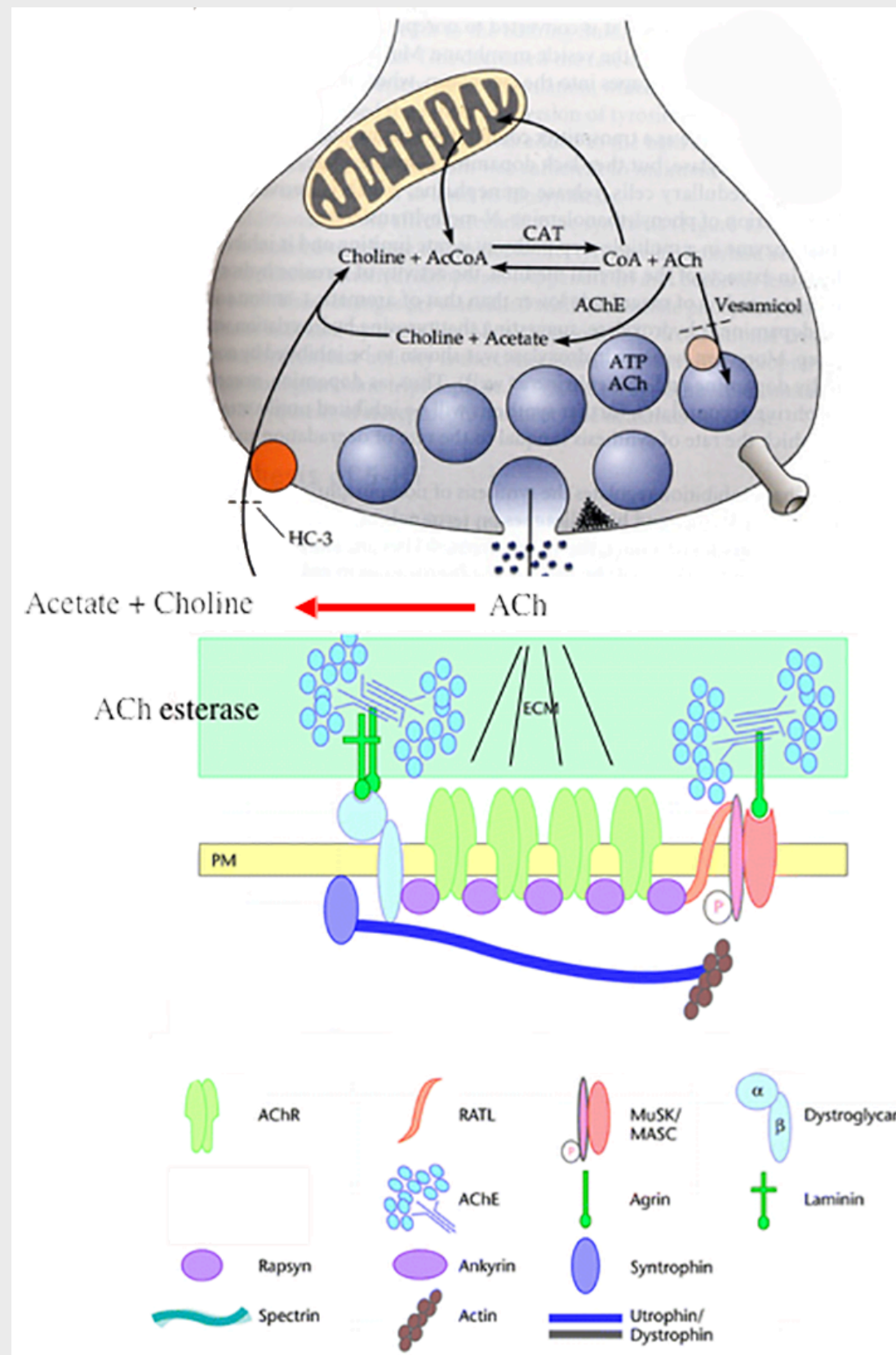
RAZ



the neurotransmitter must be quickly removed from the synaptic cleft

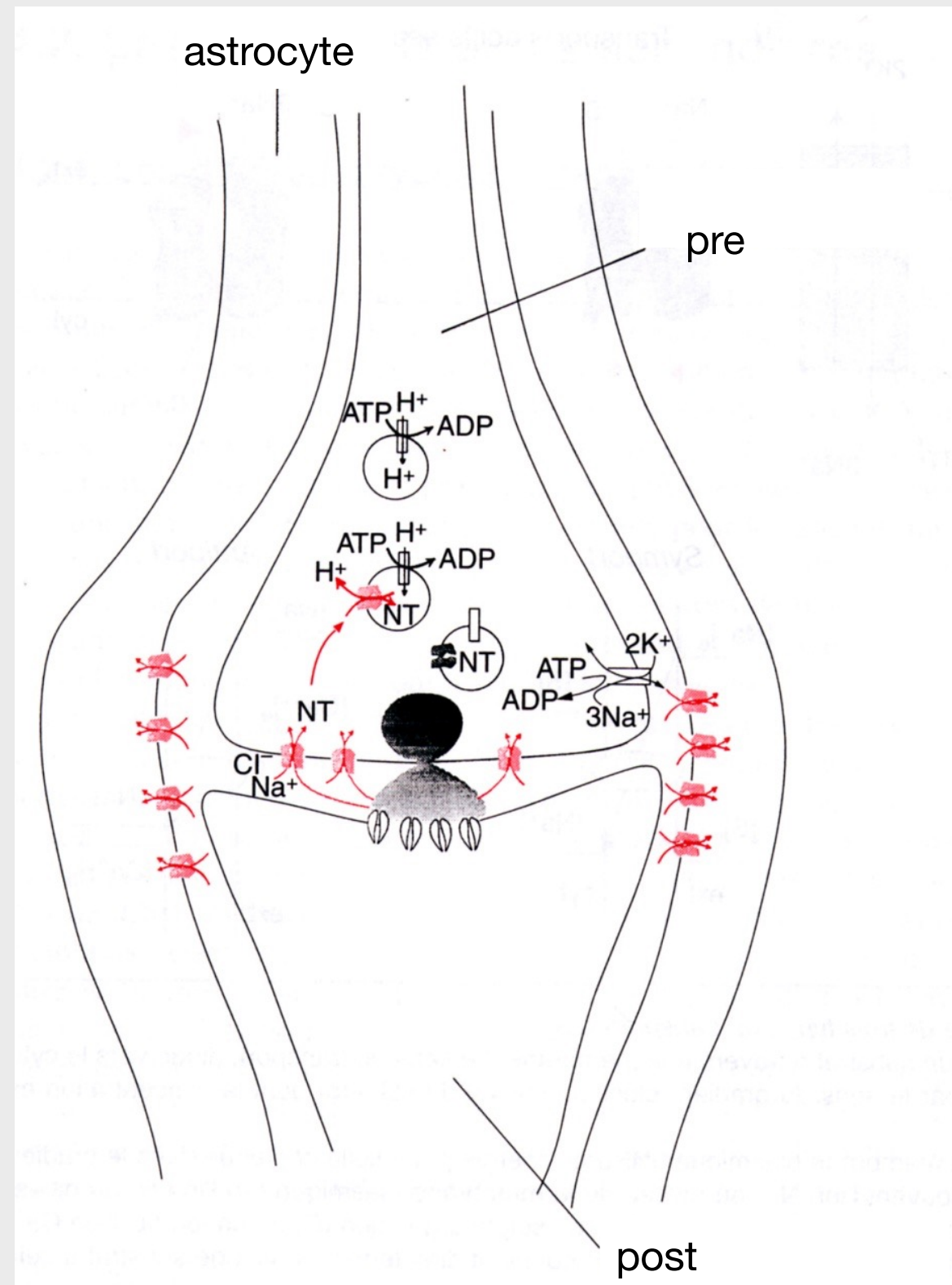
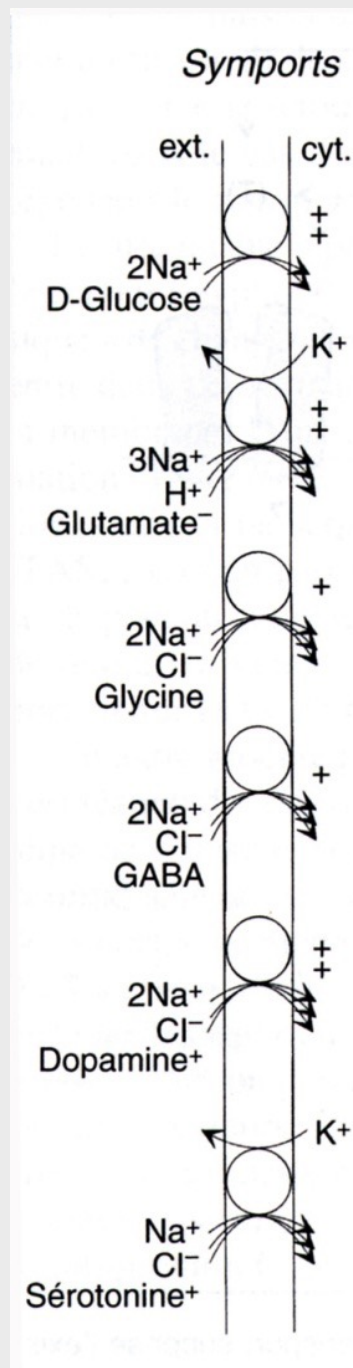
destruction/removal

RAZ

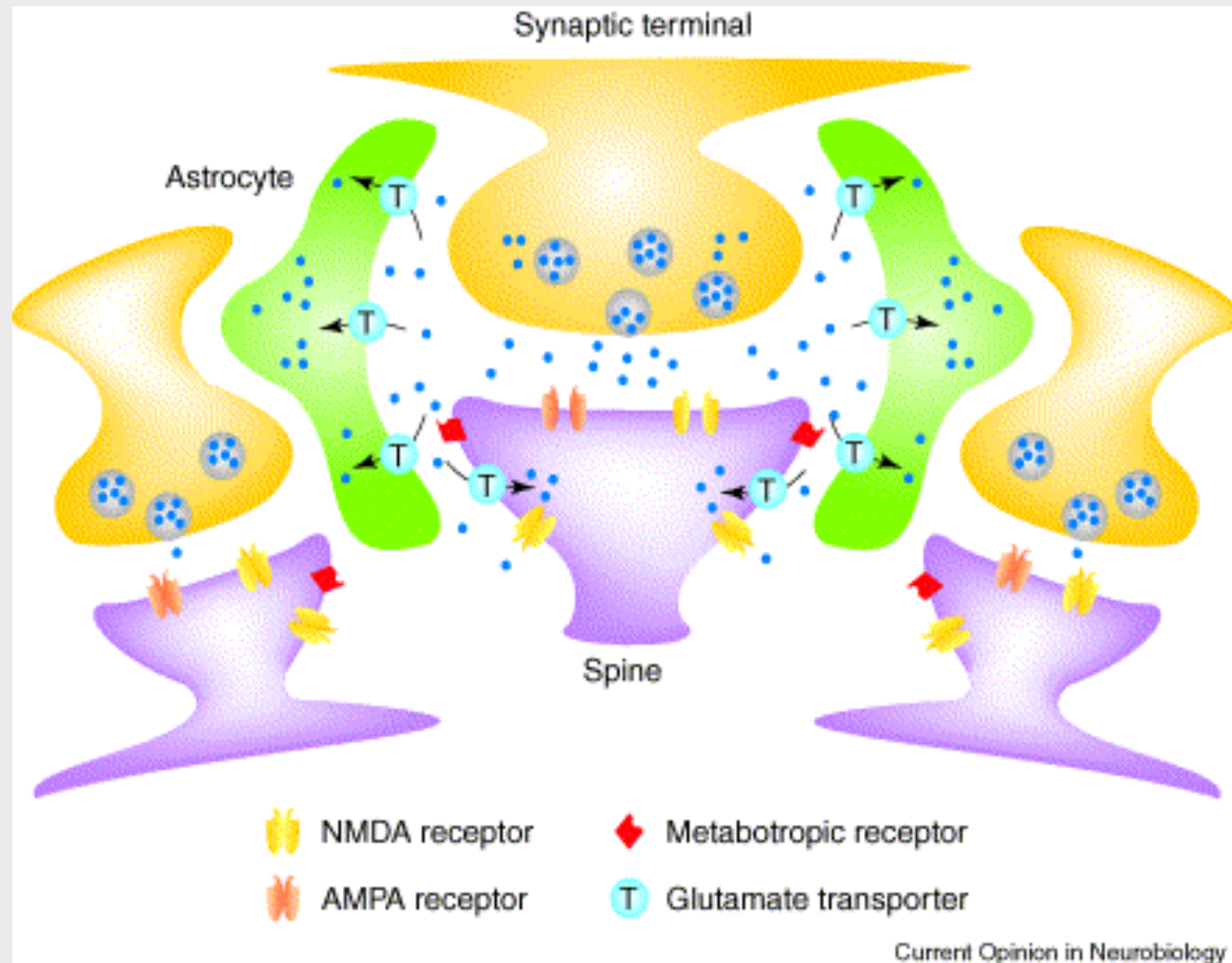


RAZ

Na⁺ gradient

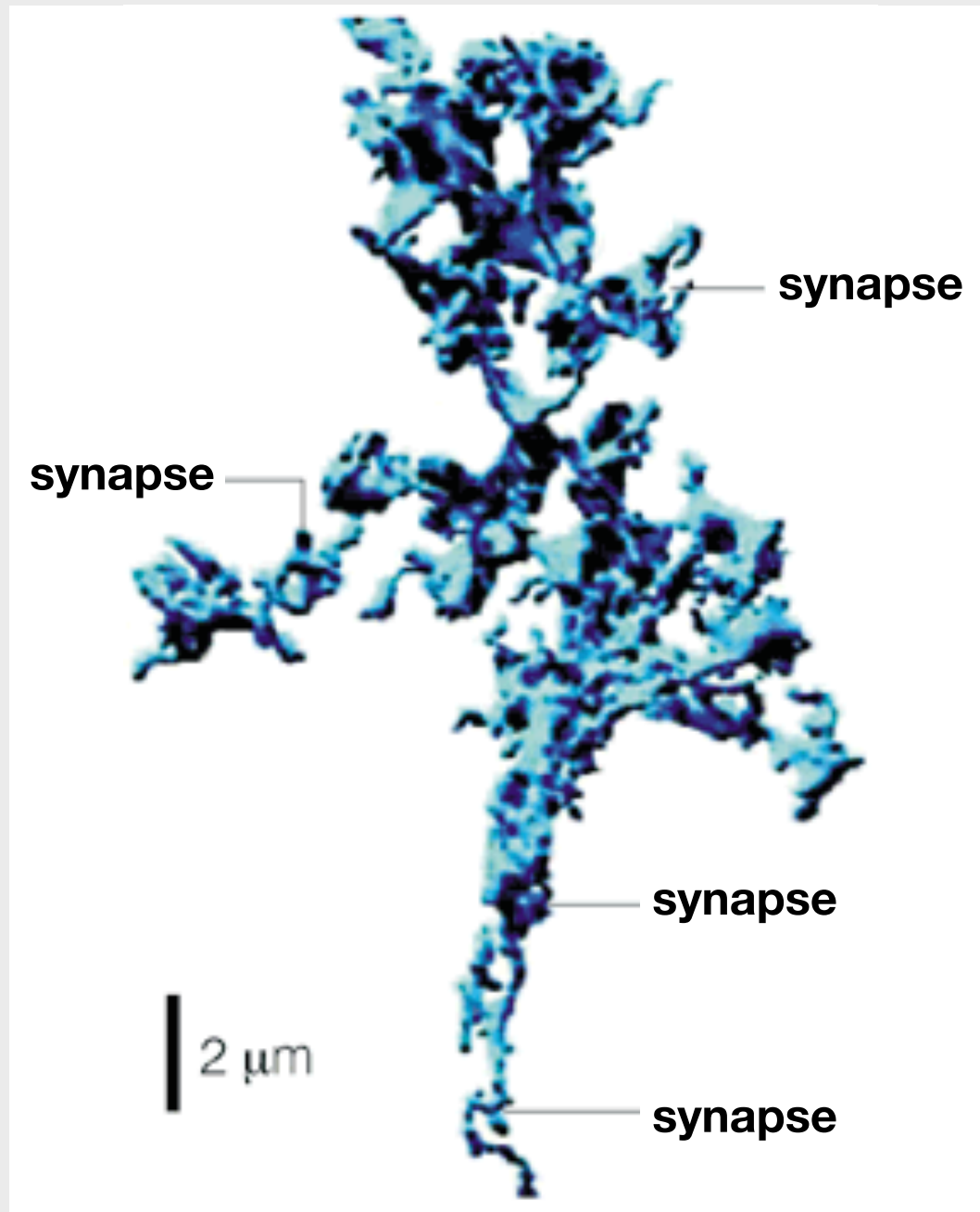


RAZ

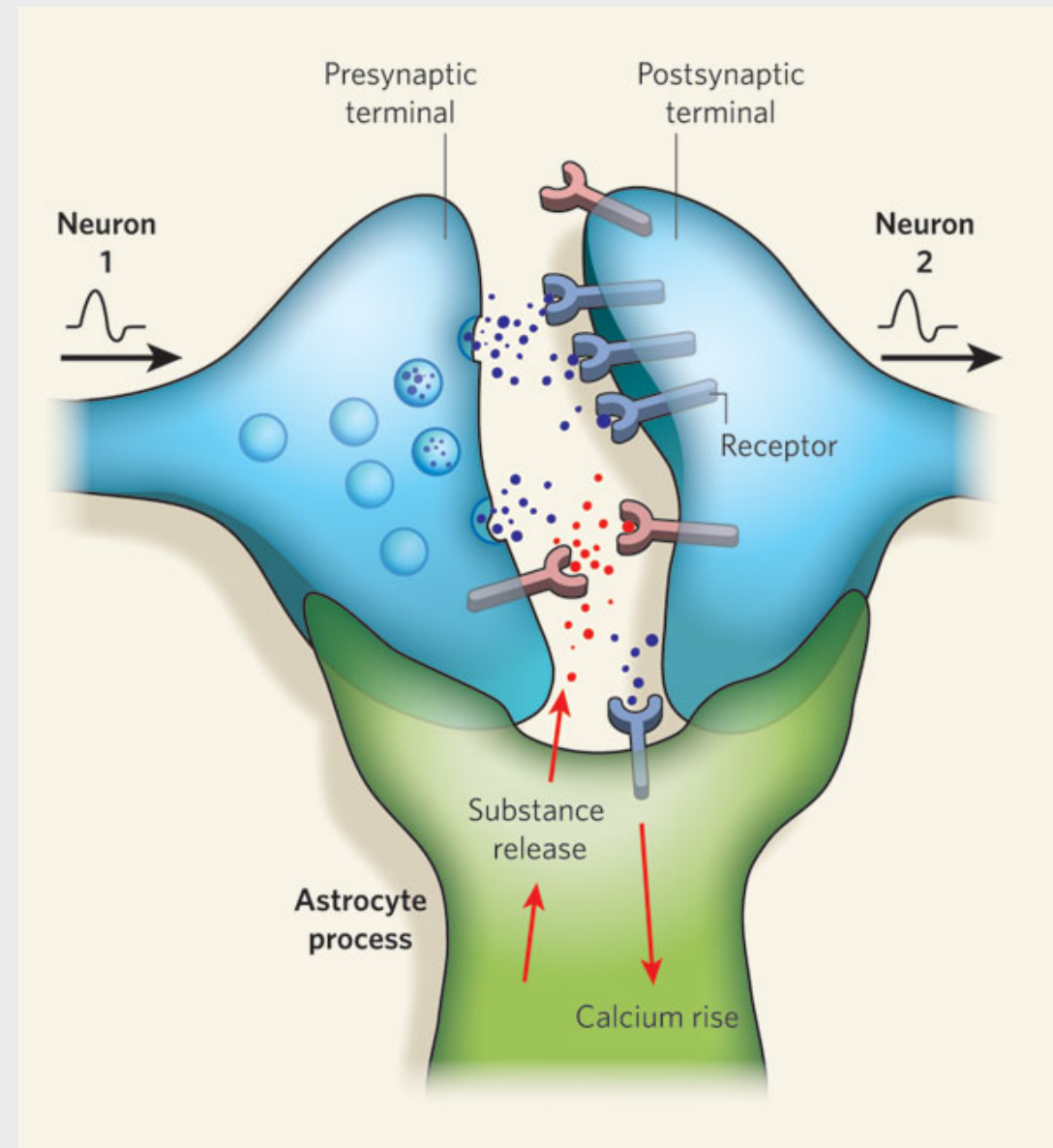


the balance between the amount of neurotransmitter released and the efficiency of recapture by the presynaptic element and astrocytes determines whether receptors outside the active zone will be activated and whether the neurotransmitter will eventually spread to neighbouring synapses (spillover)

tripartite synapse

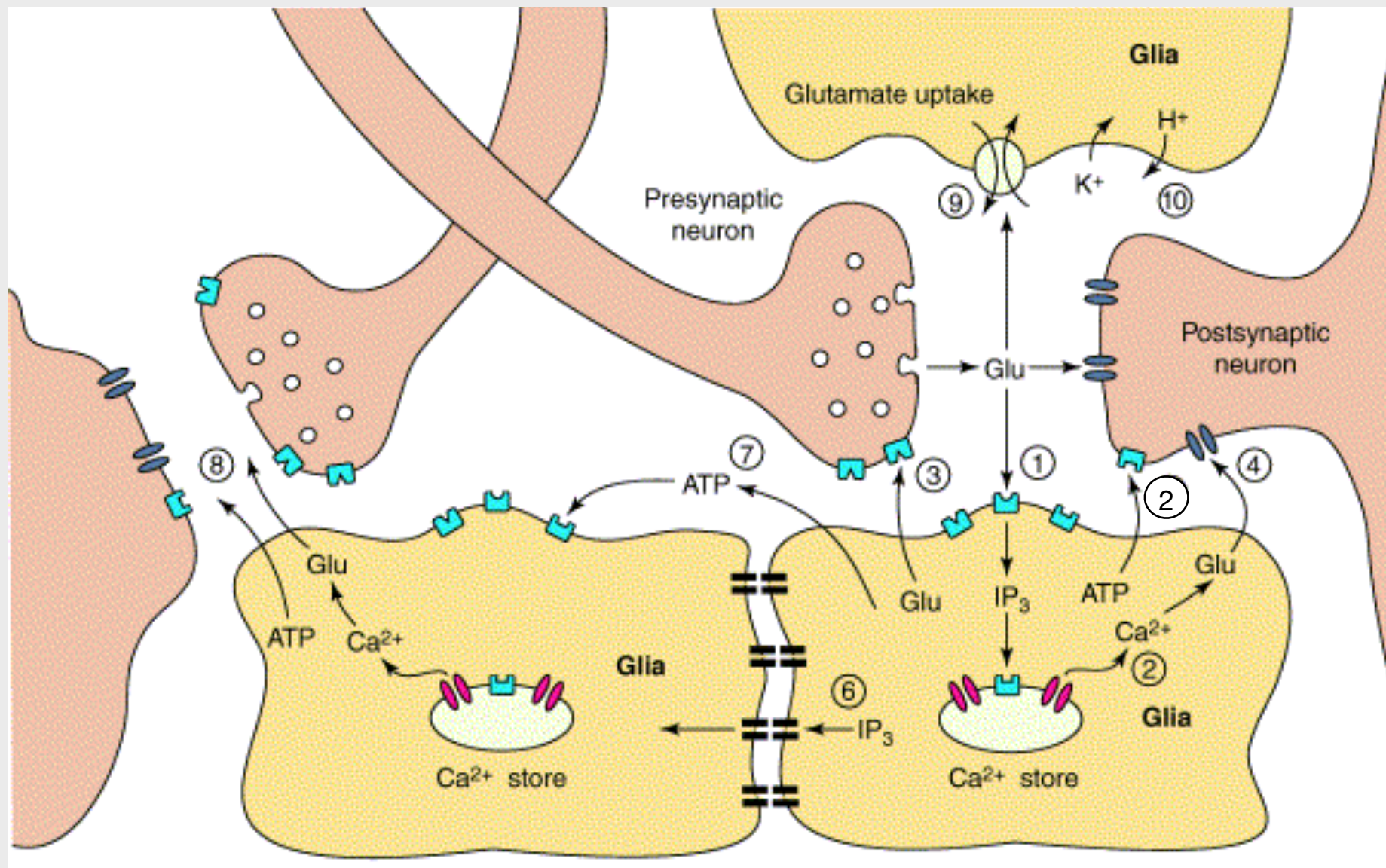


astrocytes express metabotropic receptors



gliotransmitters : glutamate, ATP, adenosine, d-serin, cytokines, peptides ...

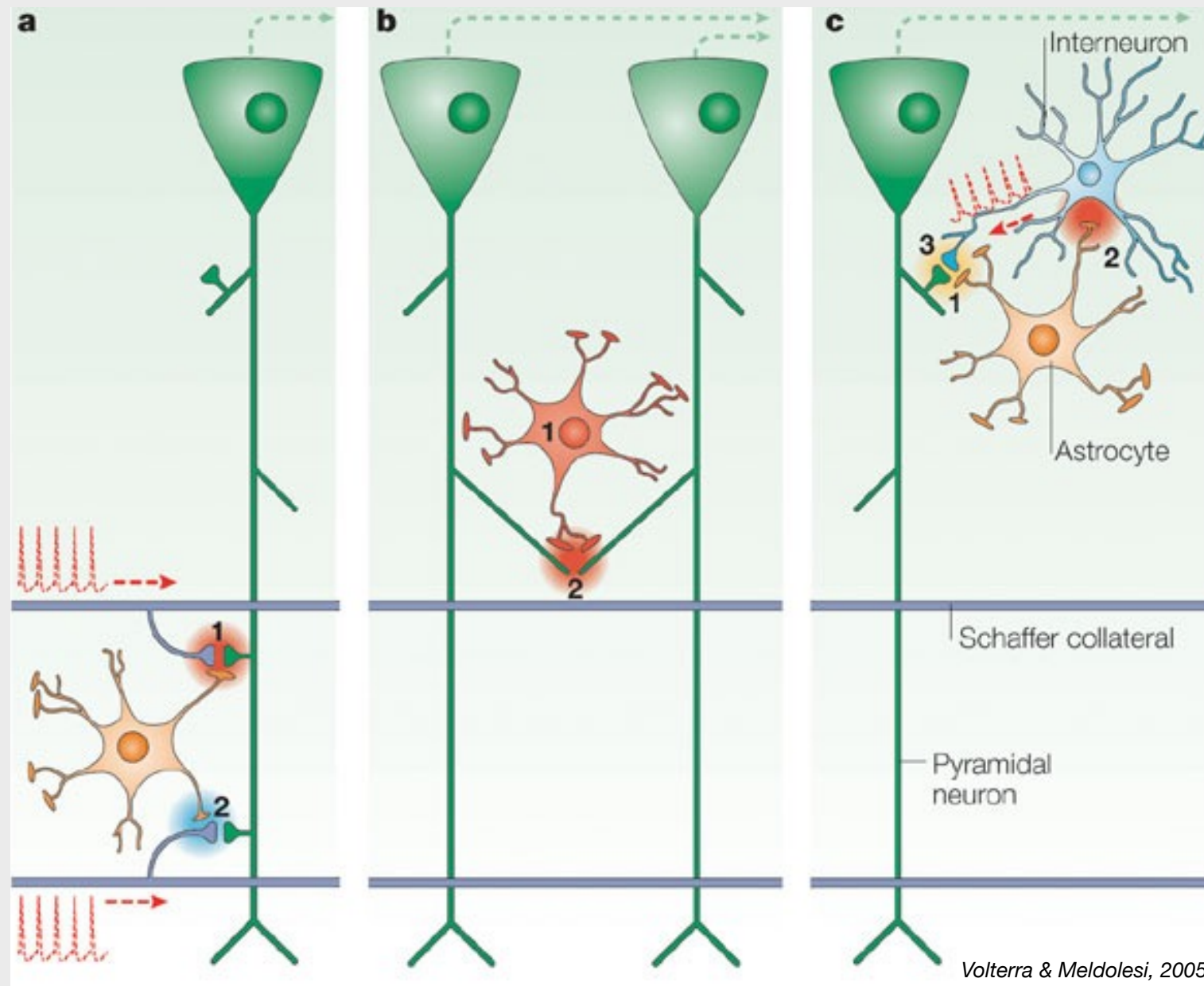
tripartite synapse



- 1 glutamate (Glu) binds to astrocyte receptors
- 2 increase of Ca²⁺ and release of gliotransmitter (Glu, ATP,...)
- 3 action on presynaptic receptors (control of neurotransmitter release)
- 4 action on postsynaptic receptors
- 6 Initiation of calcium signals propagating via IP₃ diffusion through the communicating junctions
- 7 release of ATP that acts on adjacent astrocytes
- 8 modulation of remote synapses

tripartite synapse

hippocampus

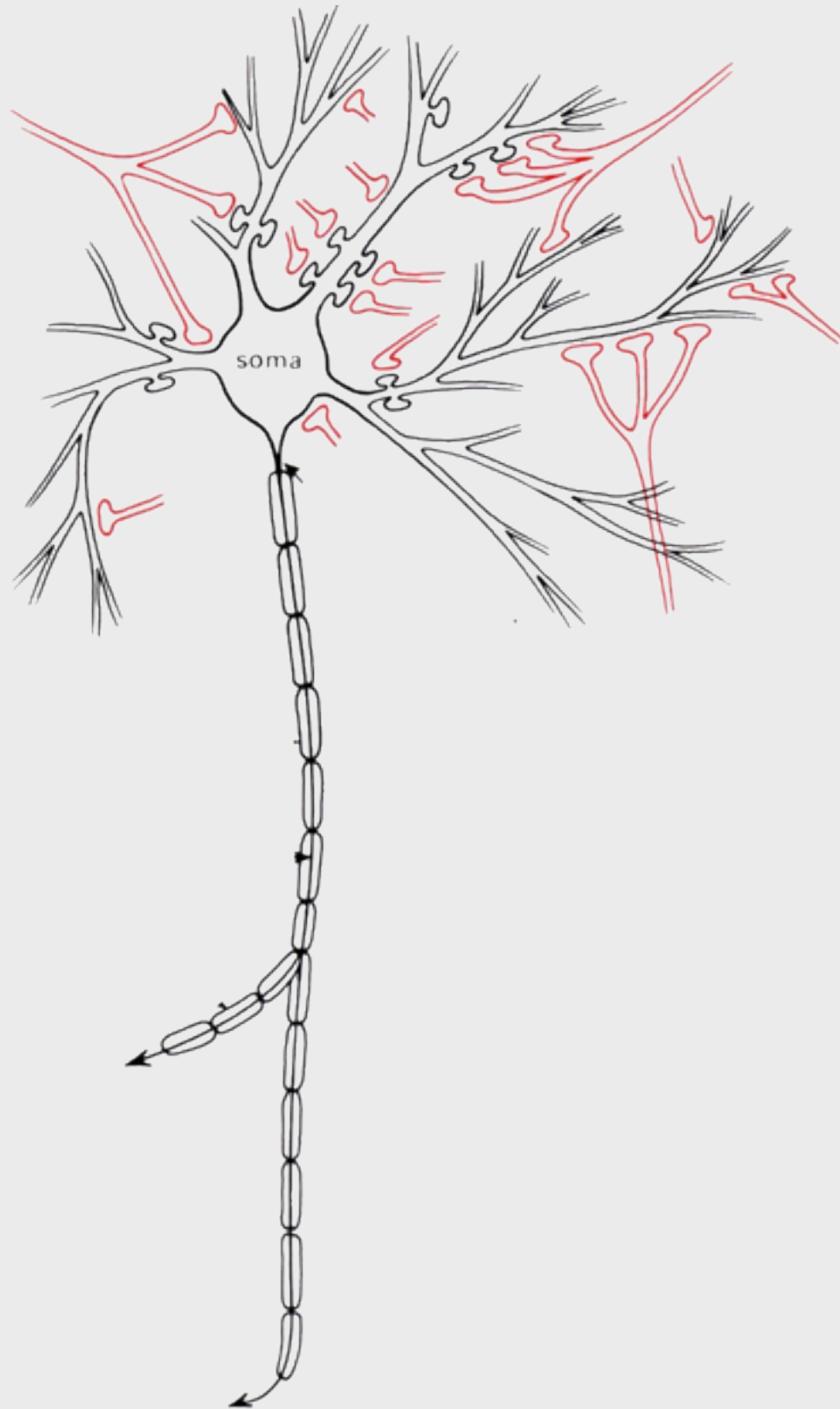


a - heterosynaptic depression: glutamate release in 1 leads to ATP release in 2 and pre-synaptic inhibition by metabotropic adenosine receptors (Zhang et al., 2003).

b - excitation and synchronisation of adjacent neurons: spontaneous oscillations of the astrocyte 1 intracellular Ca^{2+} leads to glutamate release in 2 and activation of the NMDA receptors of the 2 adjacent neurons (Felin et al., 2004).

c - potentiation of inhibitory synapse: the release of GABA in 1 activates the metabotropic GABA receptors of the astrocyte which releases glutamate in 2 which increasing the discharge frequency of the interneuron (potentiation of inhibition 3) (Liu et al., 2004).

dendritic conduction & integration



each neuron can receive several thousand connections
(average number of synapses of a neocortical neurons : 7000)

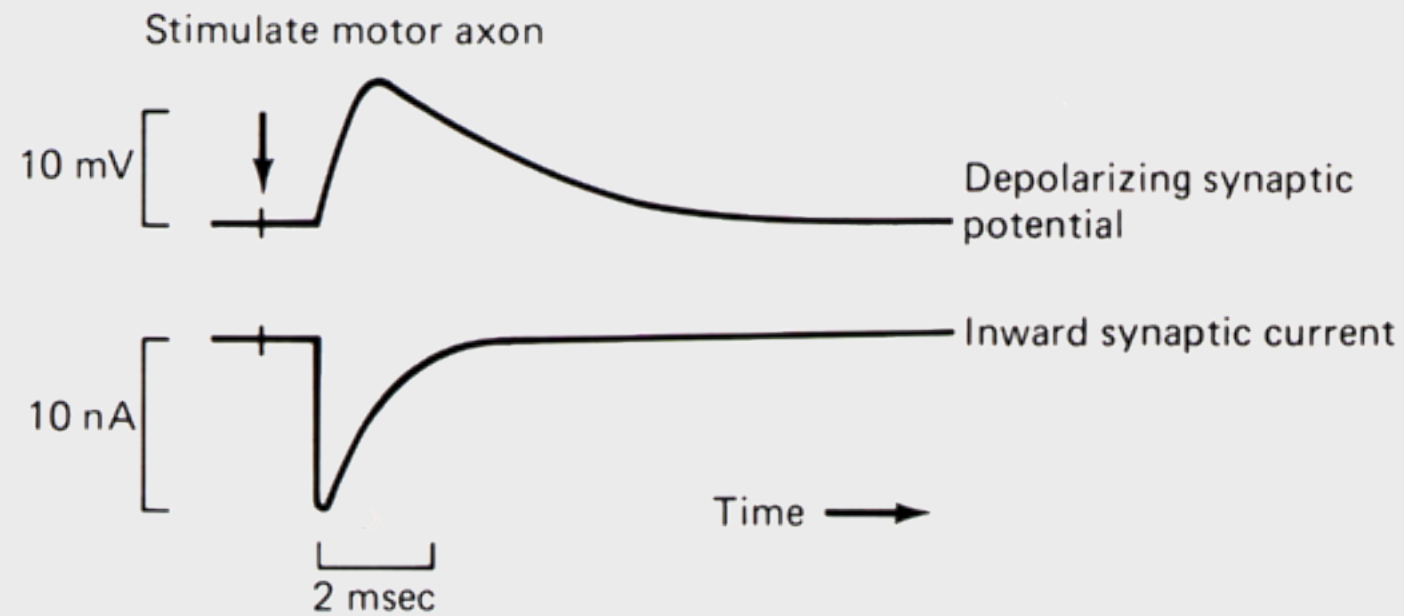
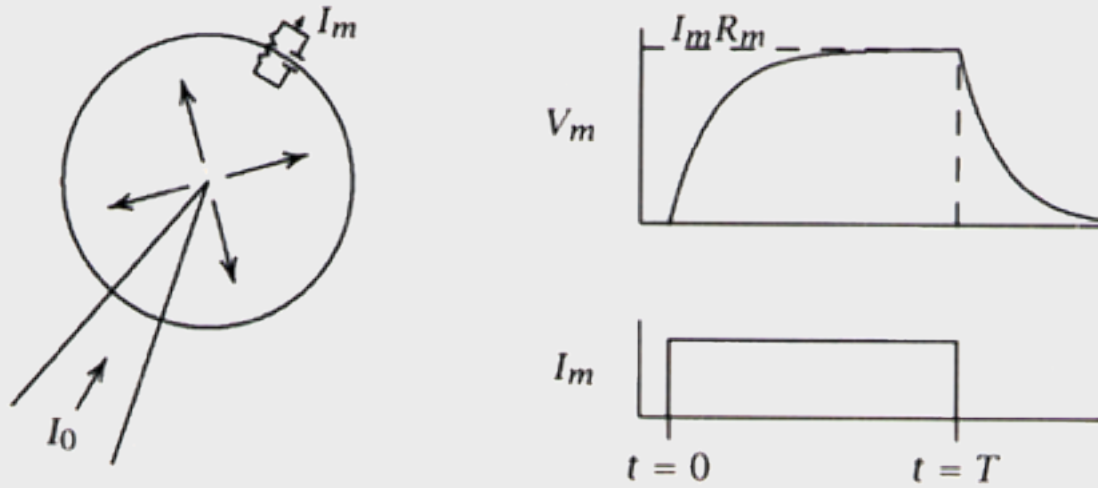
synapses are located on spines, dendritic shaft, large/
small compartments ...



- what is the local signal ?
- what is seen at the soma/AIS ?
- How do the different inputs interact in time and space ?

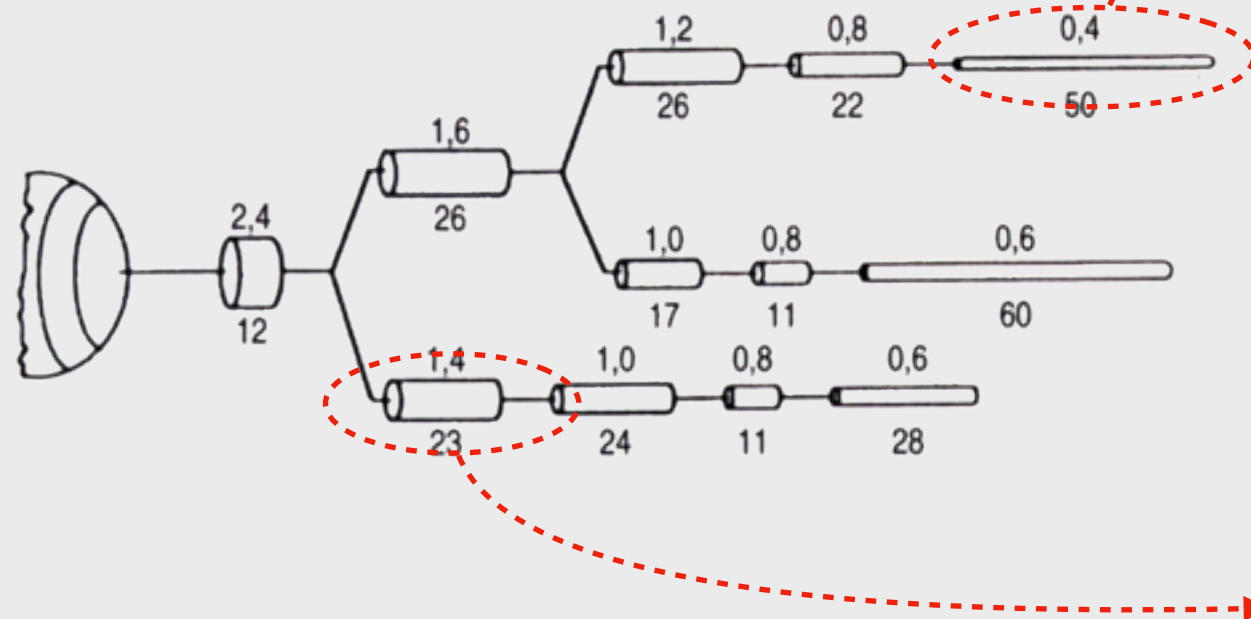
dendritic conduction & integration

potential exponentially increases with a time constant: $\tau = R_m \times C_m$.

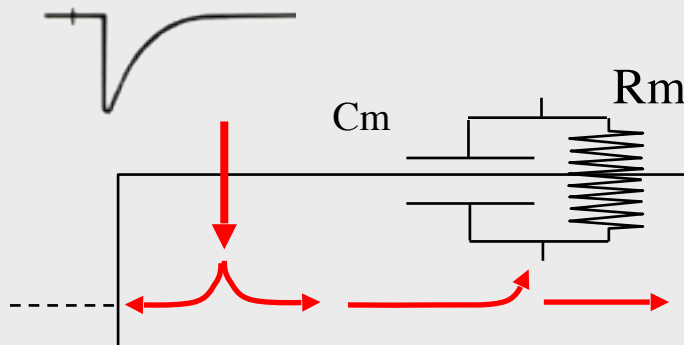


dendritic conduction & integration

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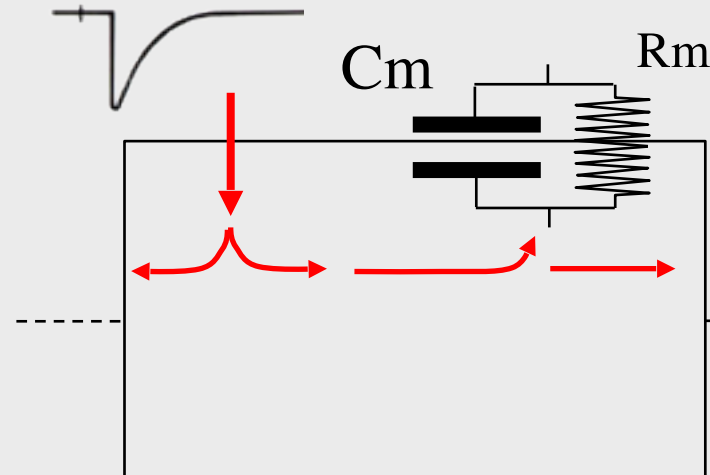
synaptic current



EPSP



synaptic current



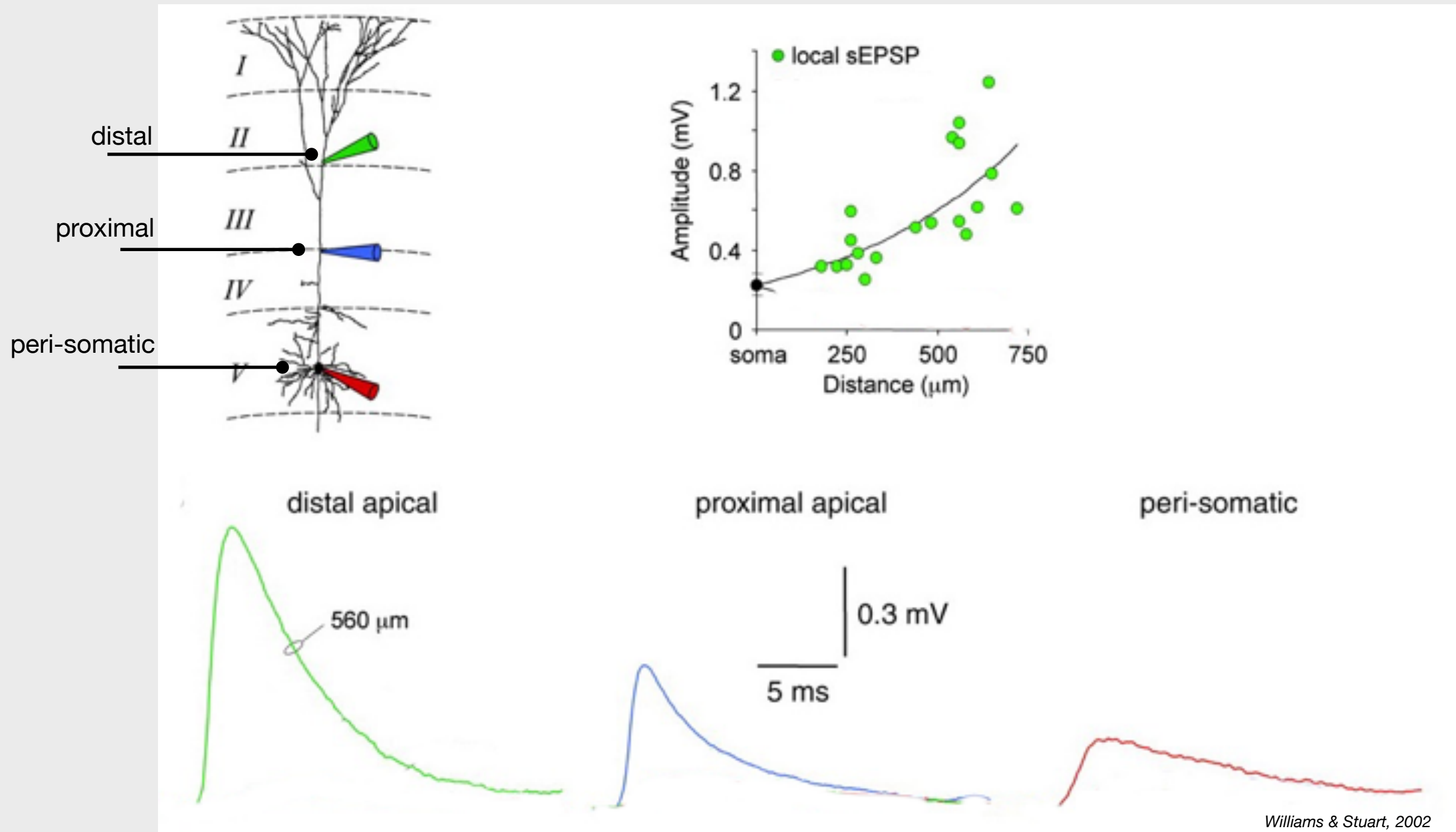
EPSP



For a given synaptic strength, the synaptic potentials will be greater in the fine distal dendrites than in the large proximal branches.

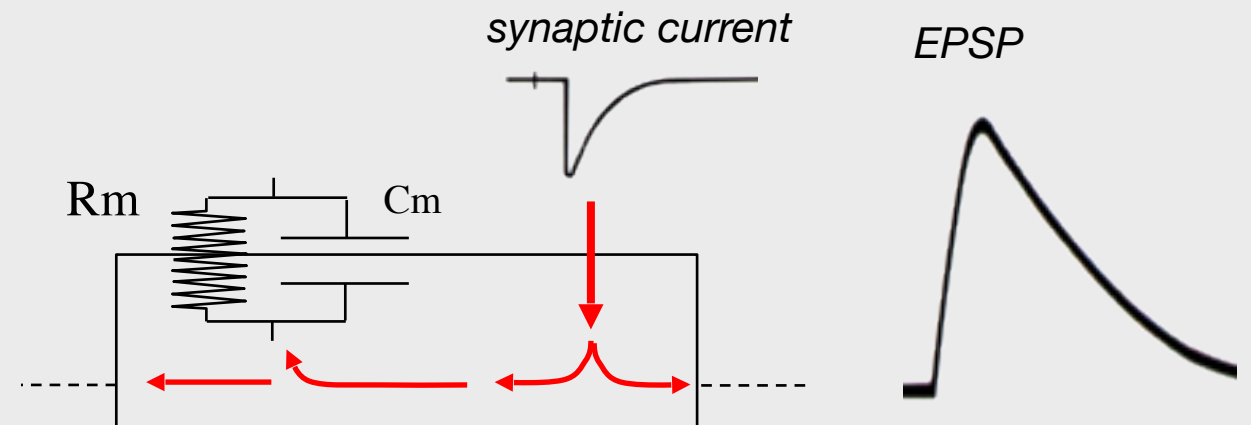
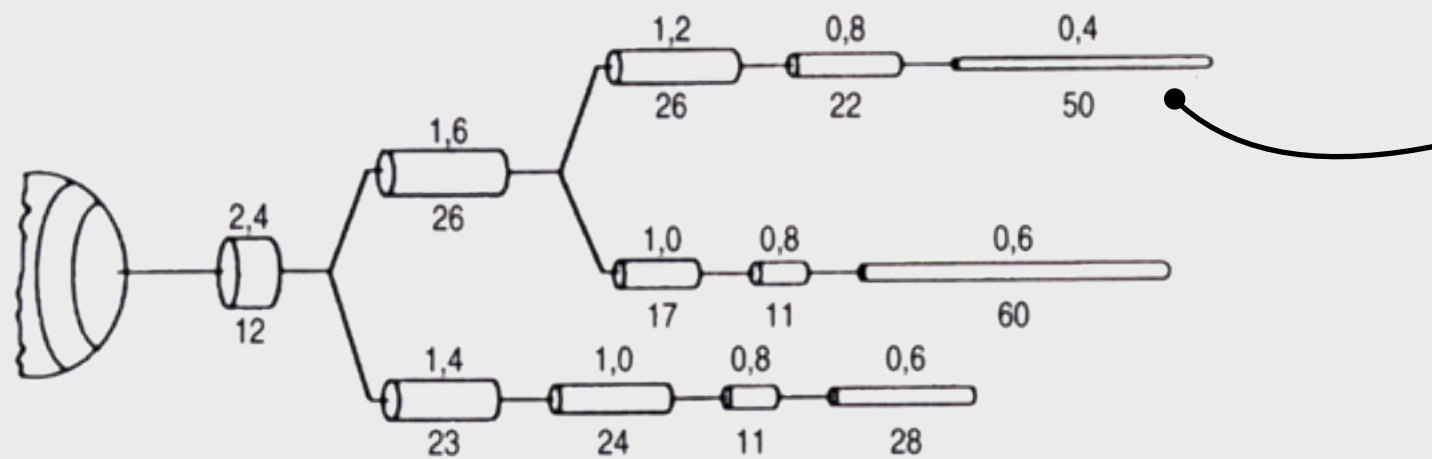
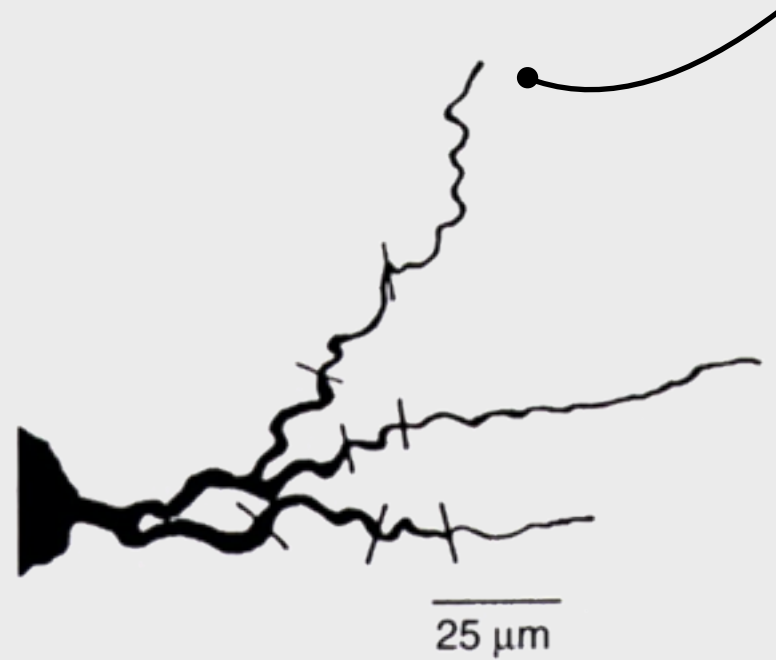
dendritic conduction & integration

cortical pyramidal cell

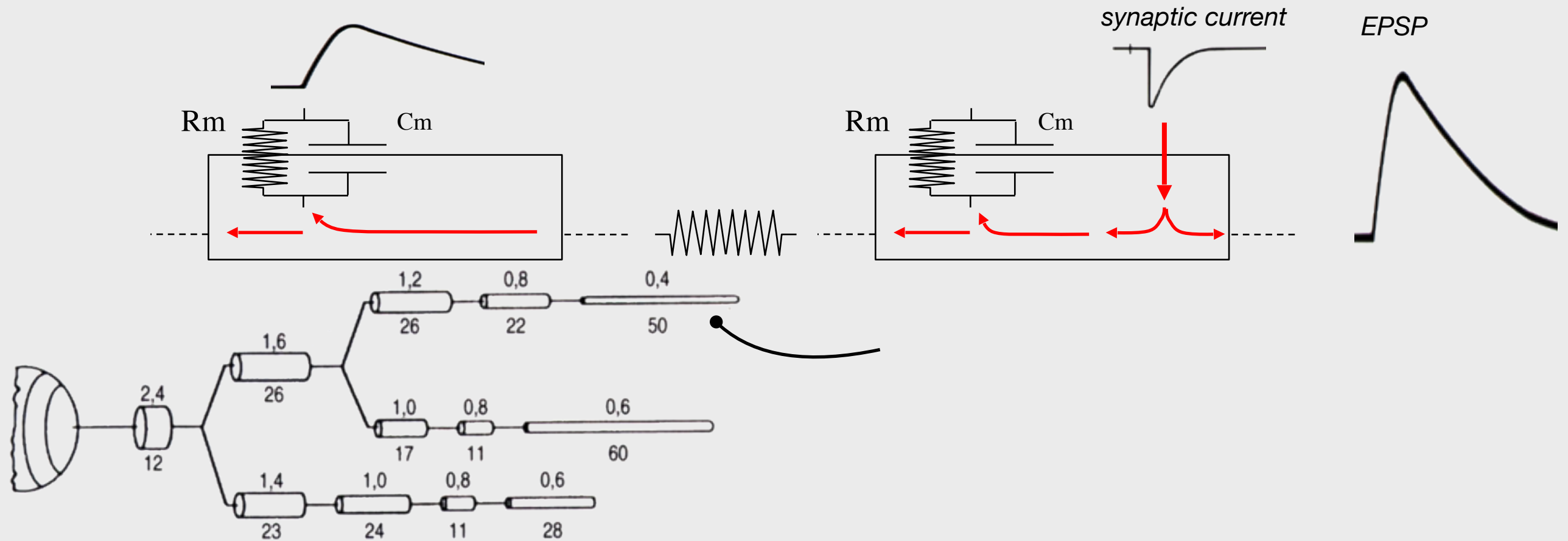


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dendritic conduction & integration

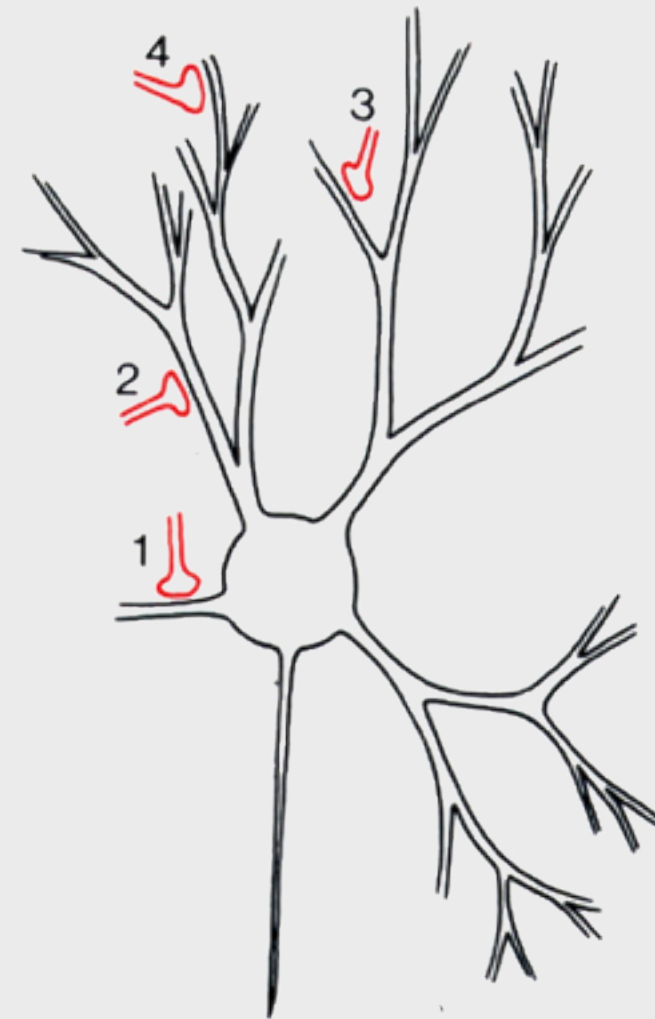
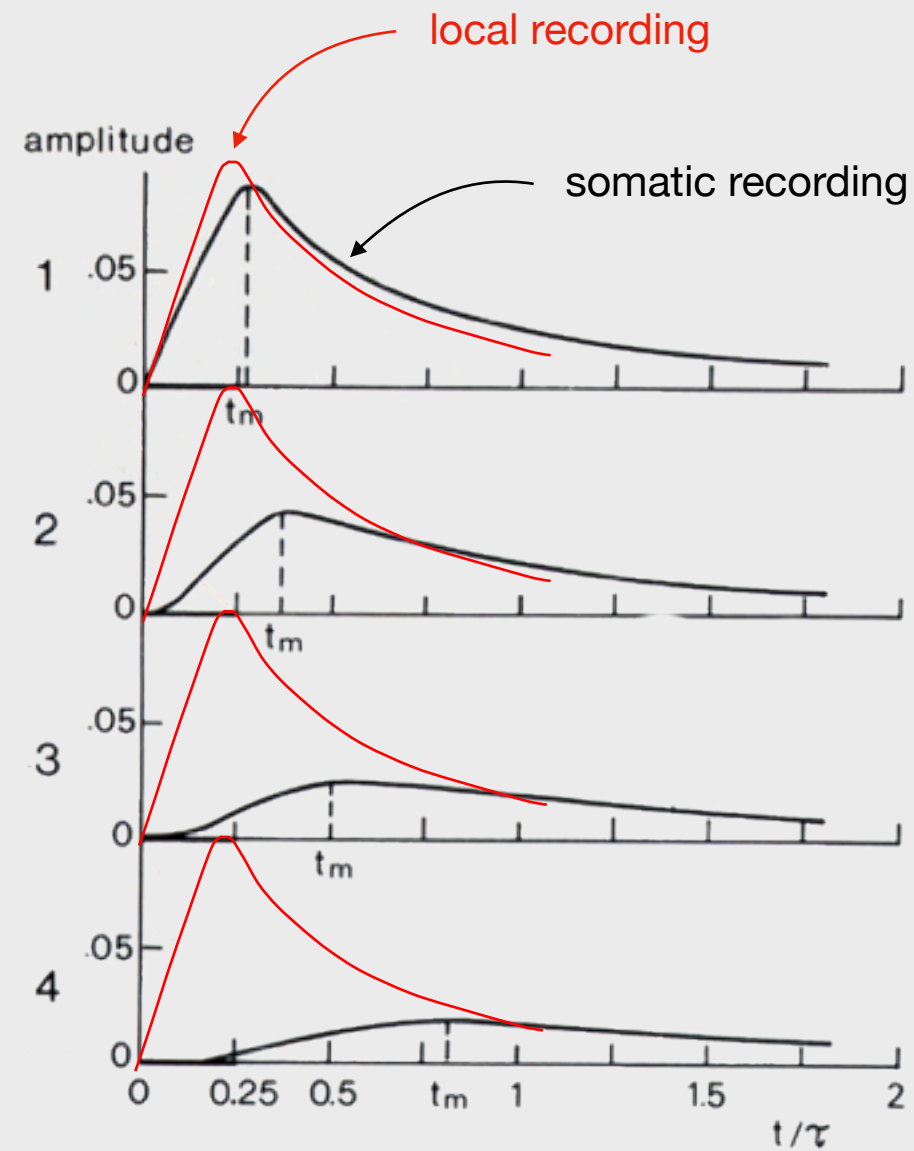


dendritic conduction & integration

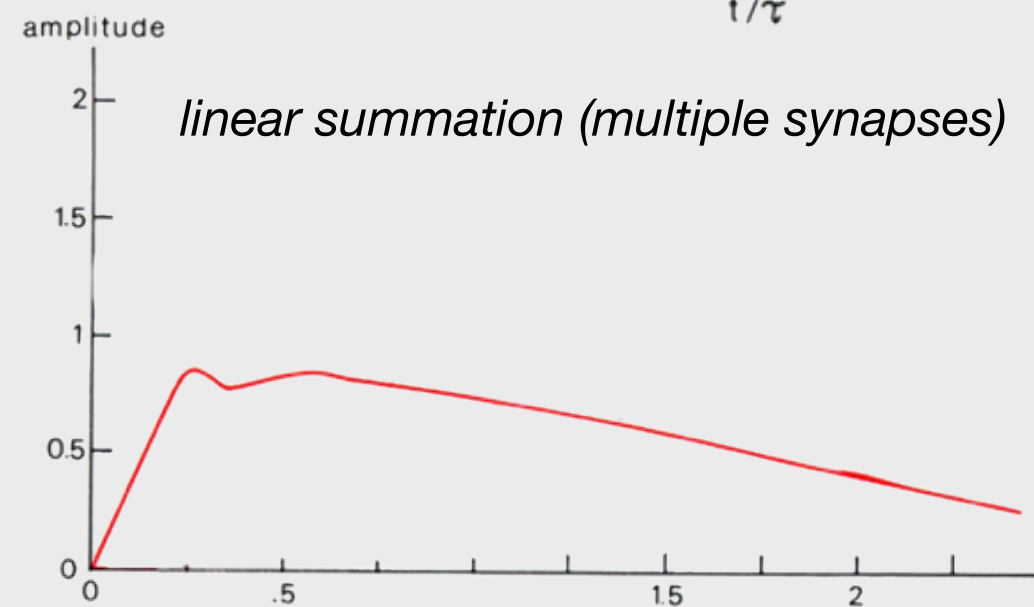
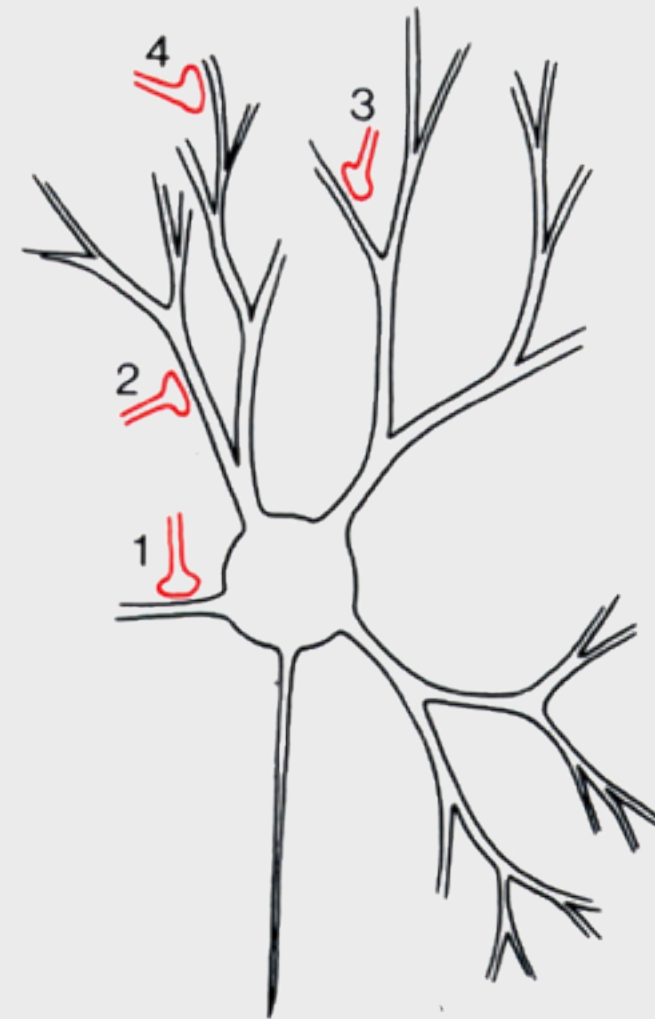
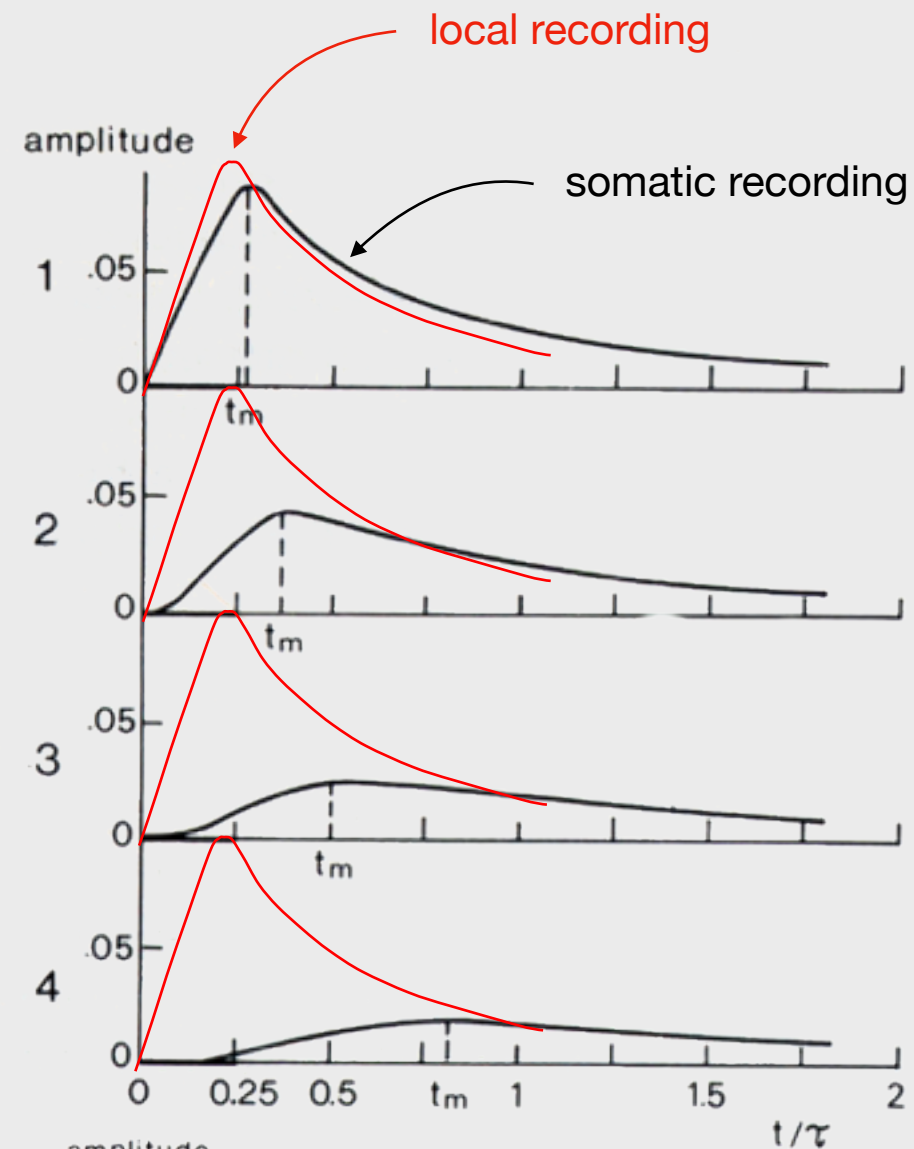


the propagation of the EPSP is associated to a progressive broadening and an attenuation in amplitude

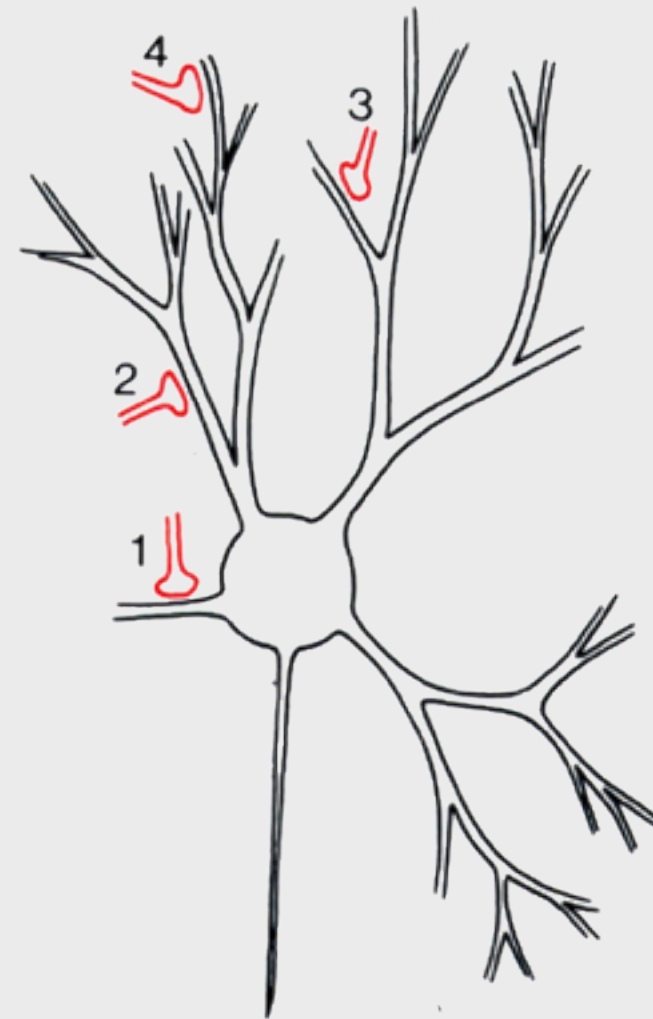
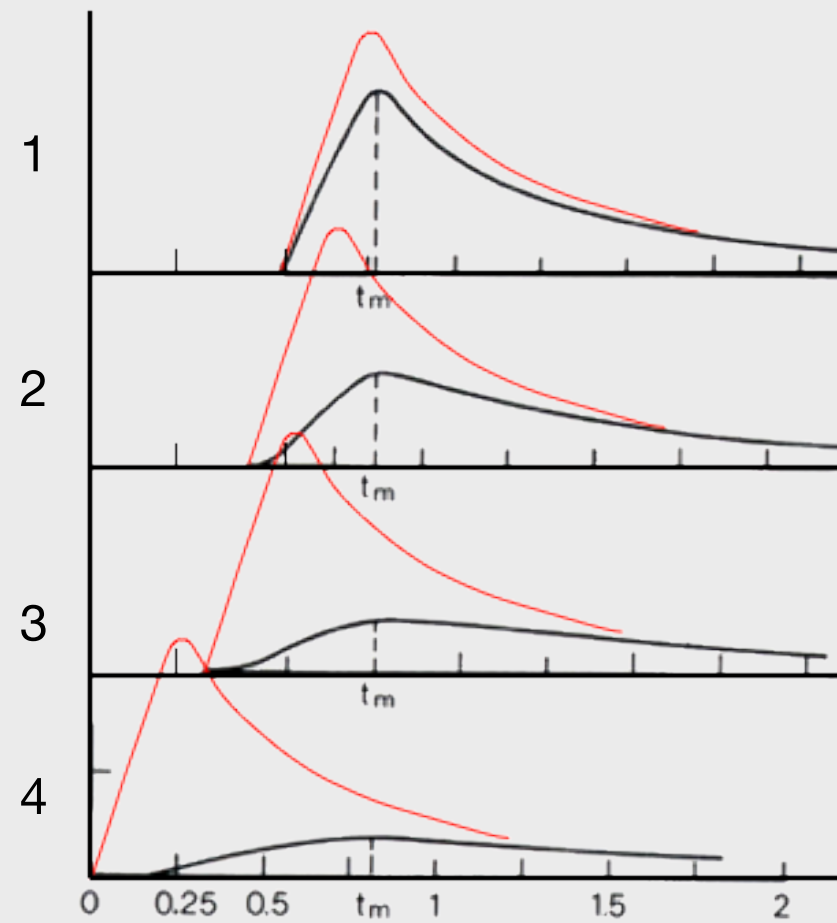
dendritic conduction & integration



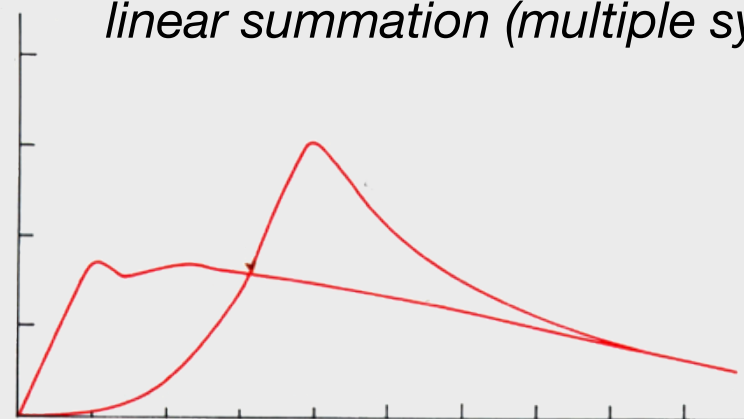
dendritic conduction & integration



dendritic conduction & integration

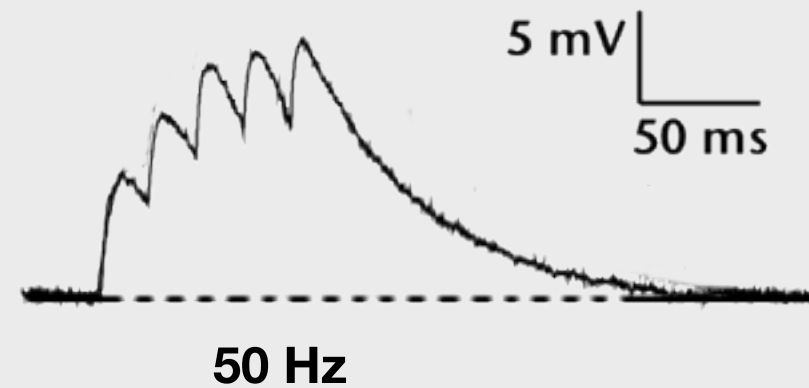
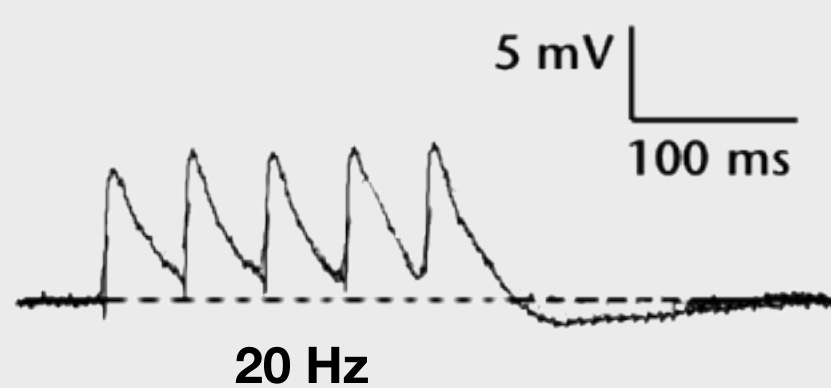
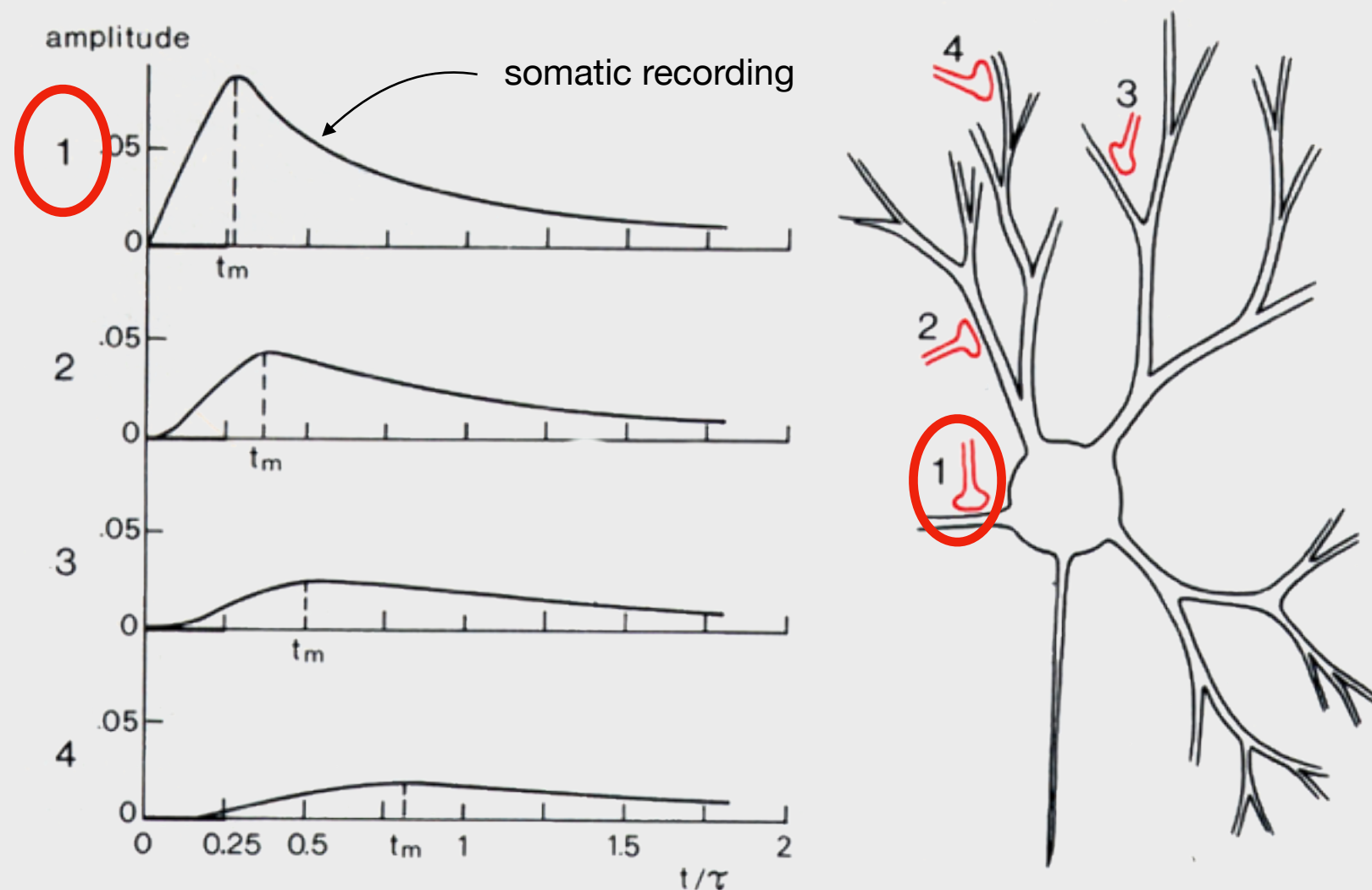


linear summation (multiple synapses)



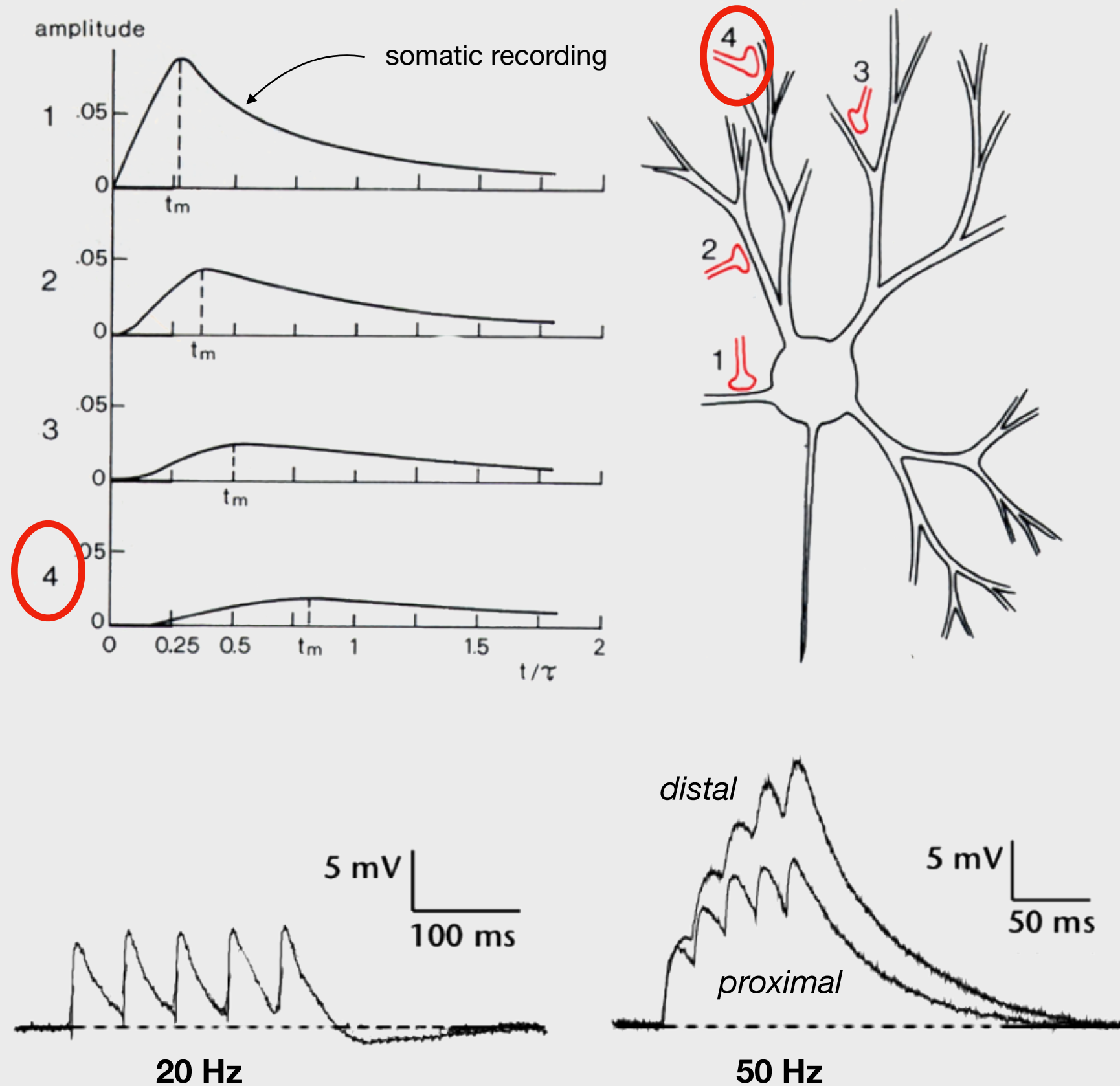
dendritic conduction & integration

temporal summation



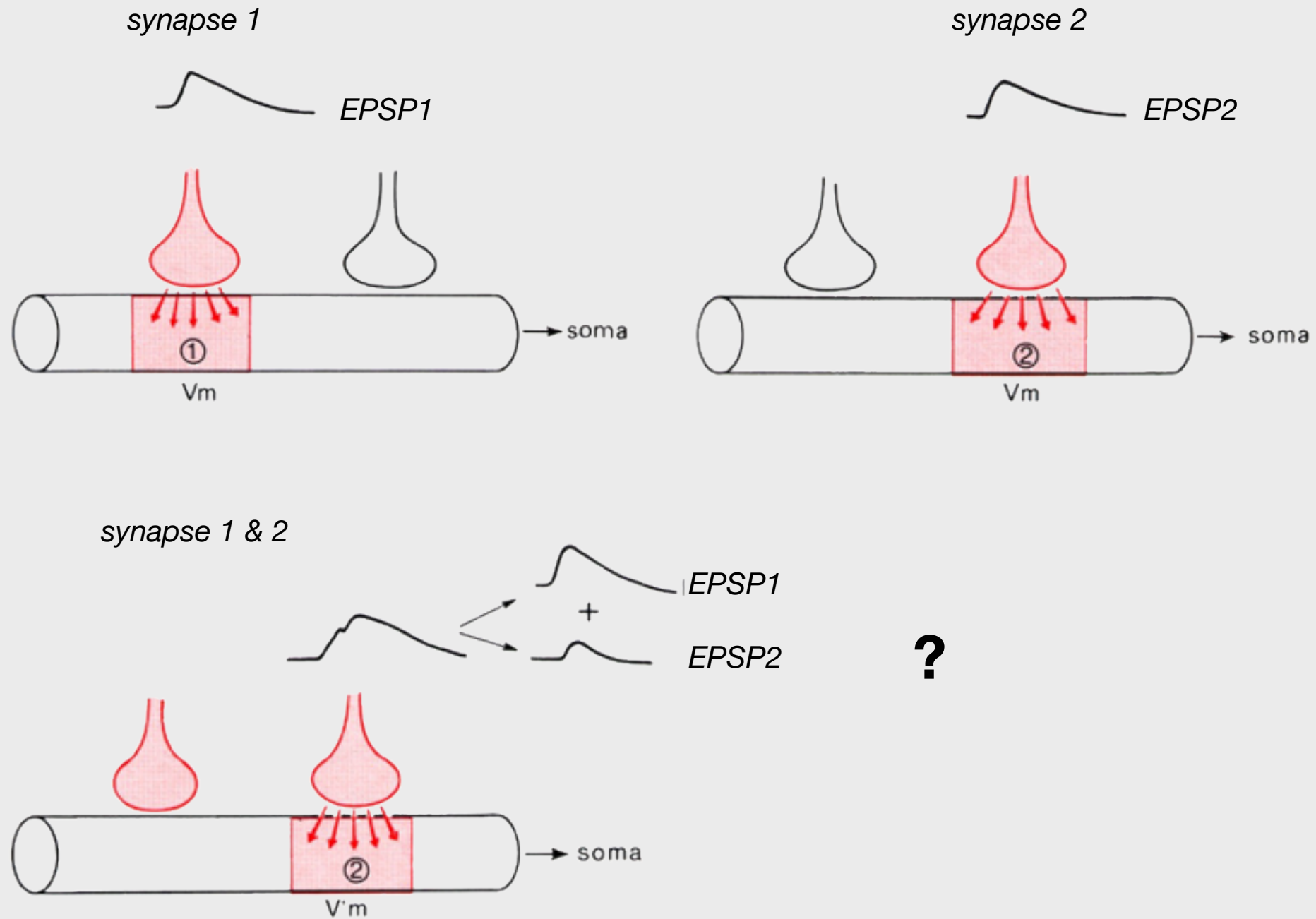
dendritic conduction & integration

temporal summation



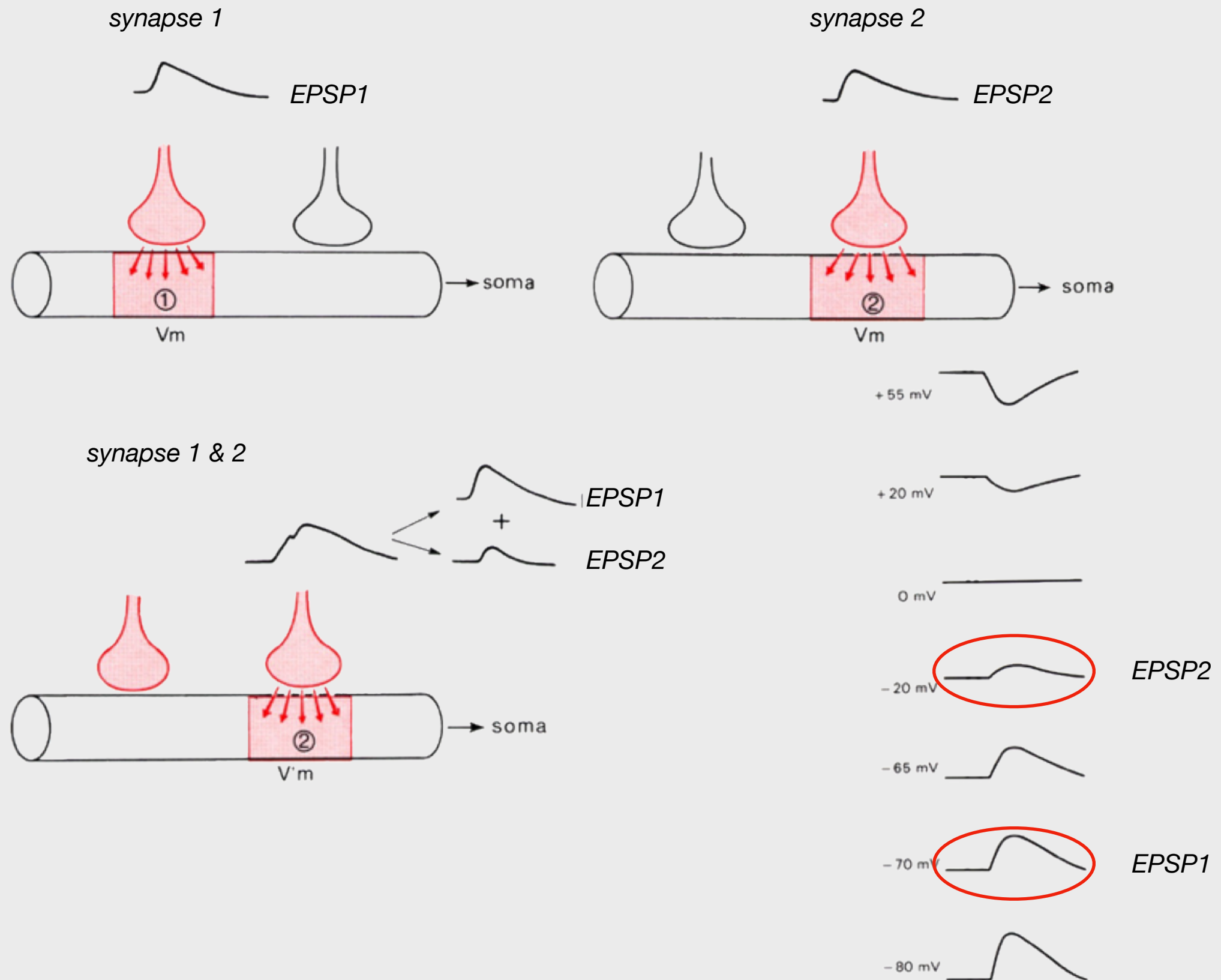
dendritic conduction & integration

non linear summation (close synapses)



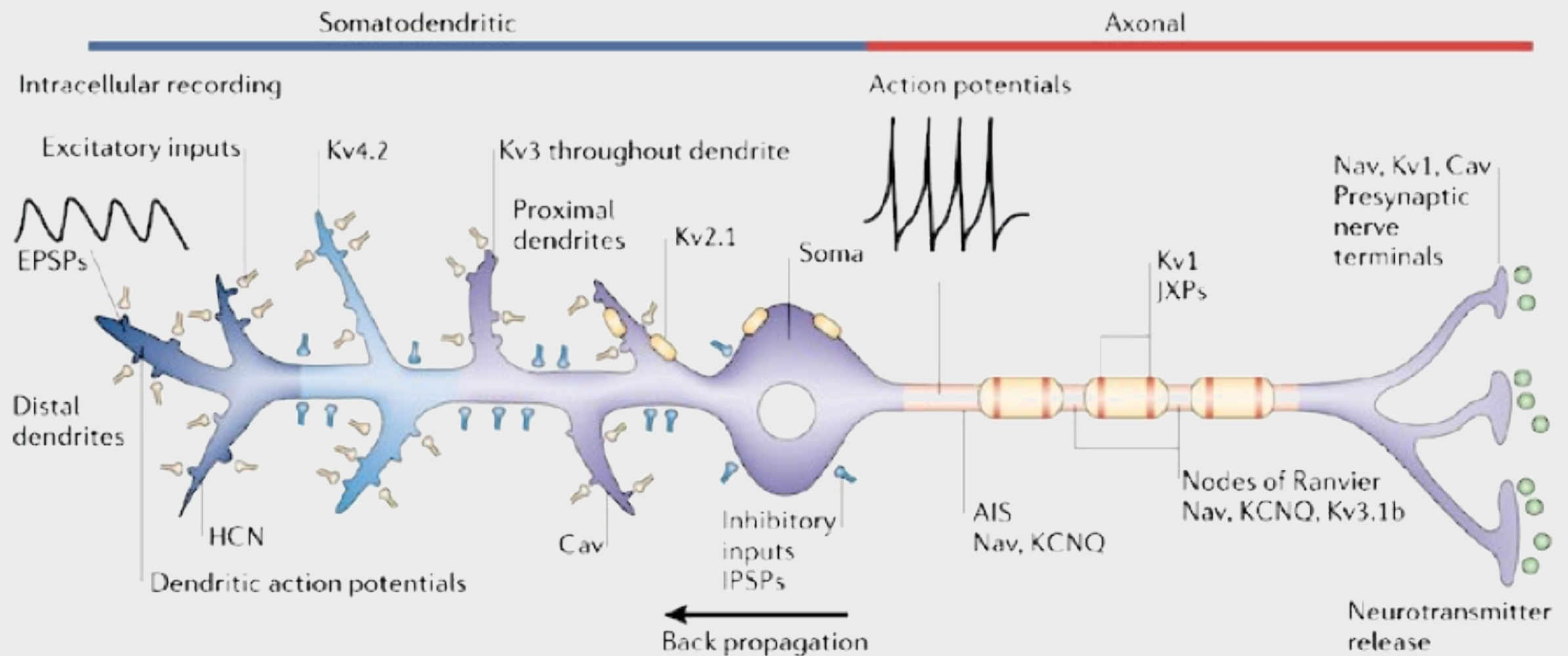
dendritic conduction & integration

non linear summation (close synapses)



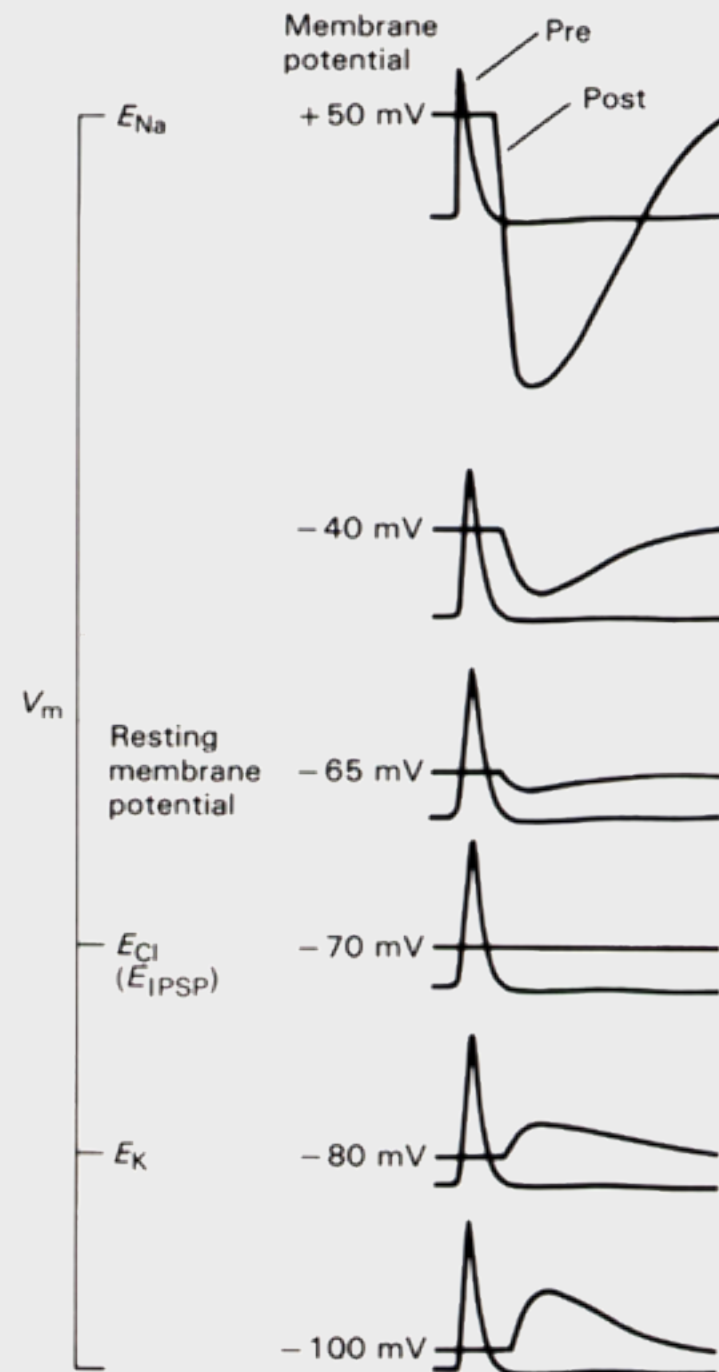
dendritic conduction & integration

*linear and non-linear summation are based on the passive properties of the dendrites and their architecture
BUT
dendrites contain many active conductances ... see you in M2 for more information*



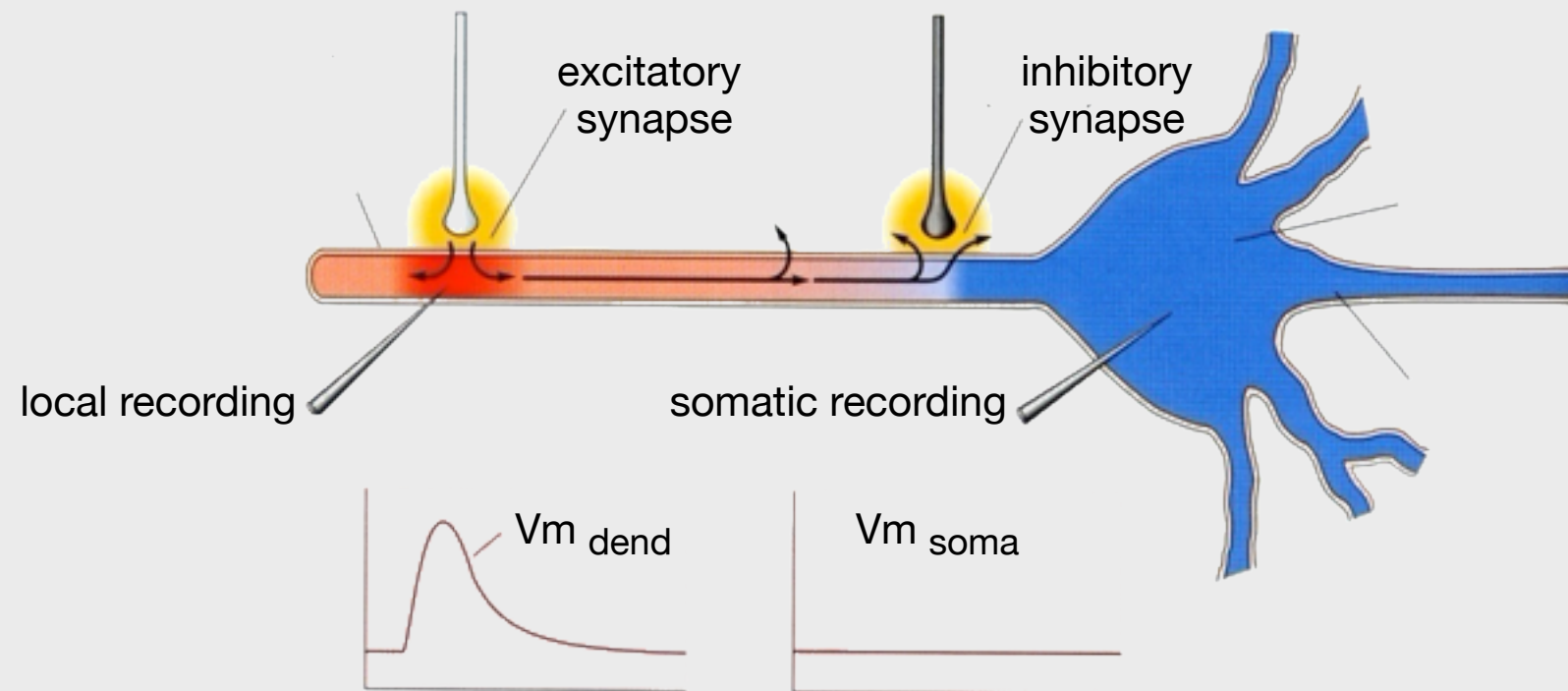
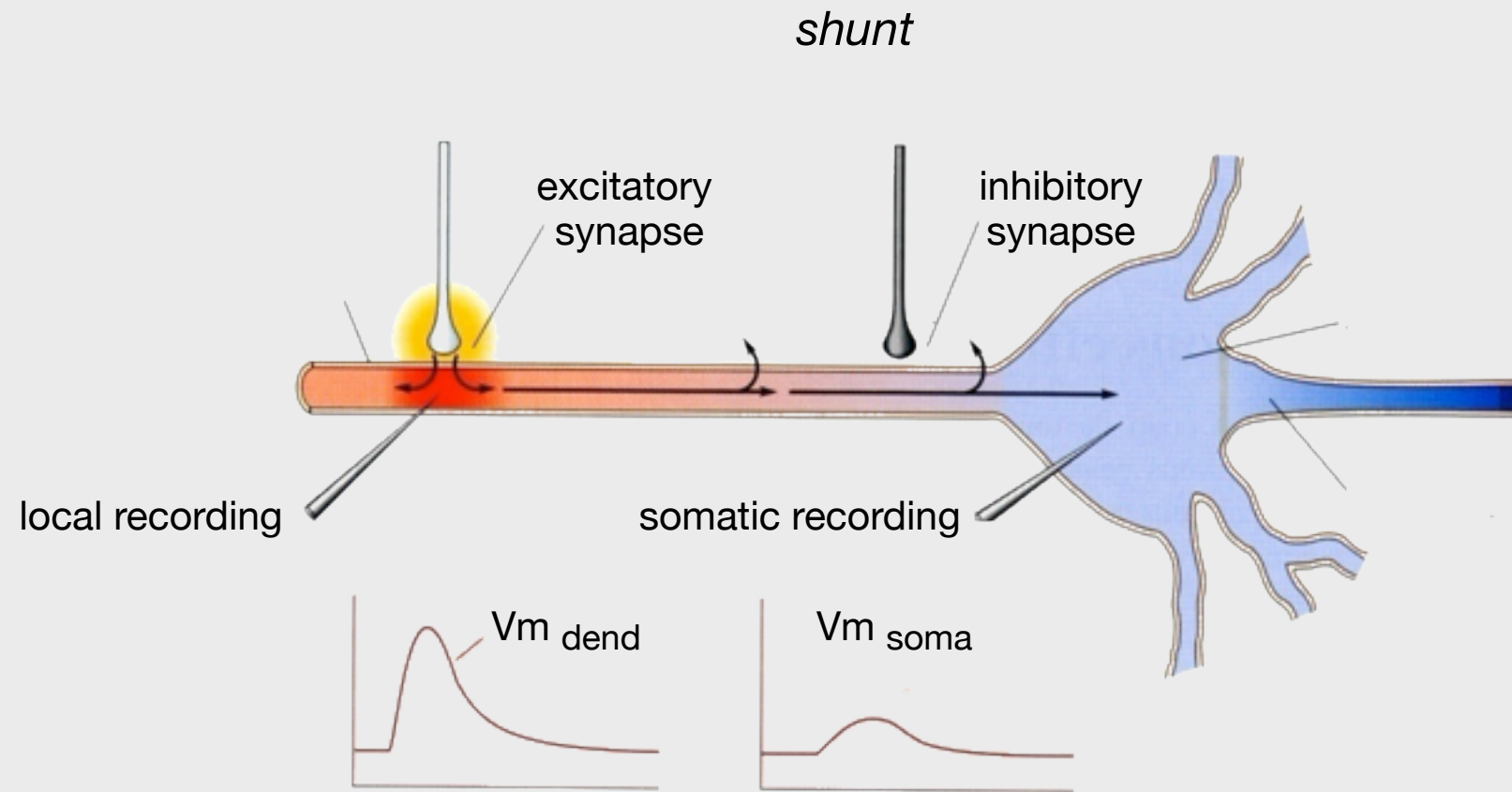
dendritic conduction & integration

the case of the inhibitory synapse



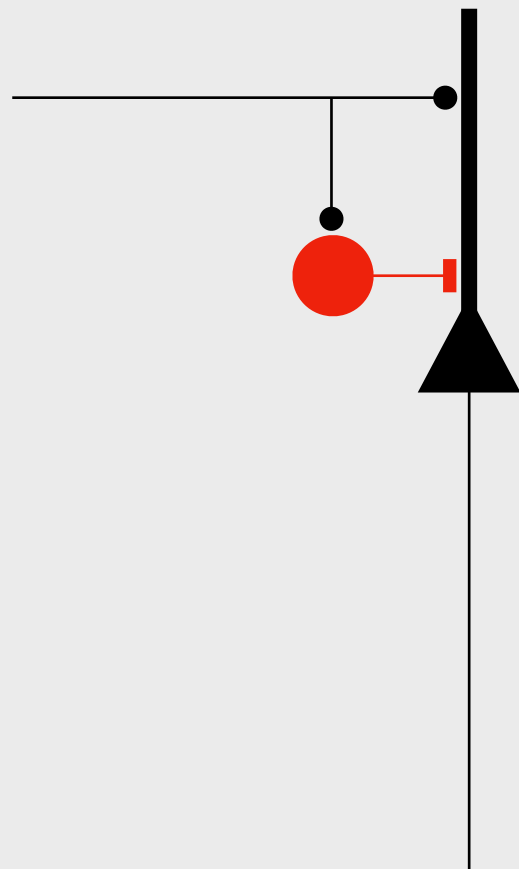
at the usual resting potential (-70 / -50 mV) activation of inhibitory synapses induces little change in the membrane potential

dendritic conduction & integration

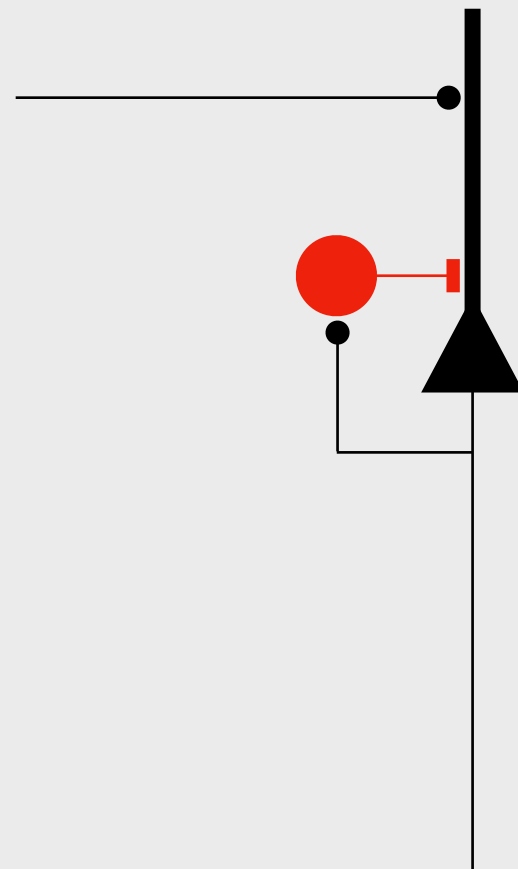


dendritic conduction & integration

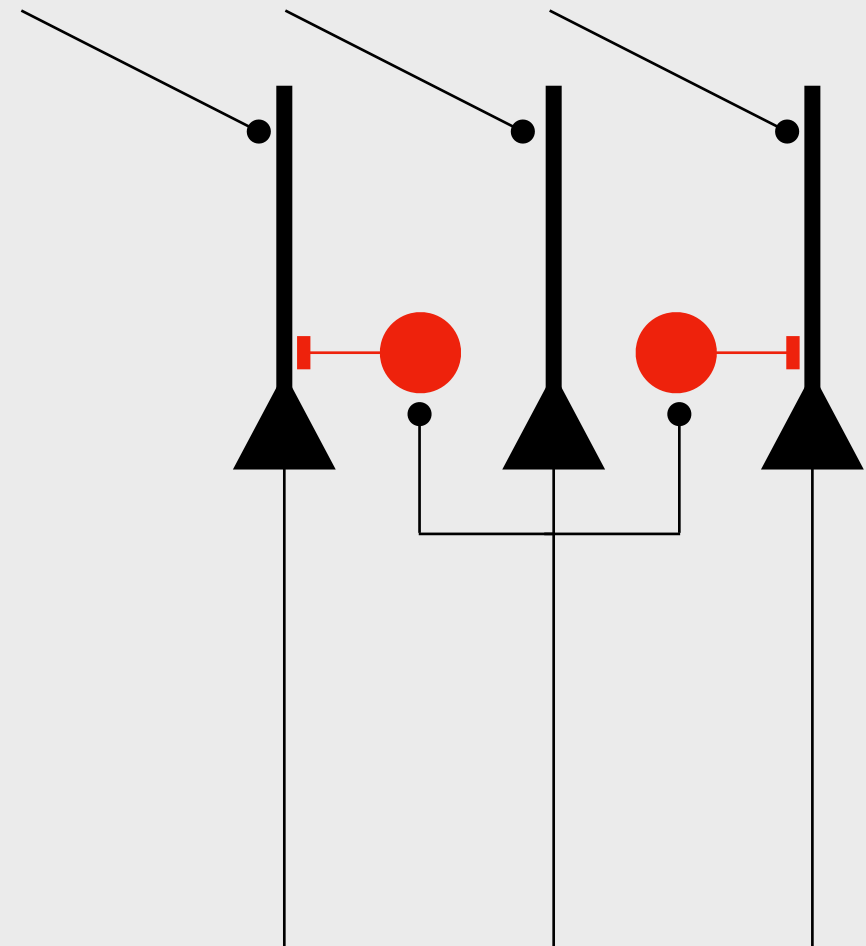
basic inhibitory circuits



feedforward



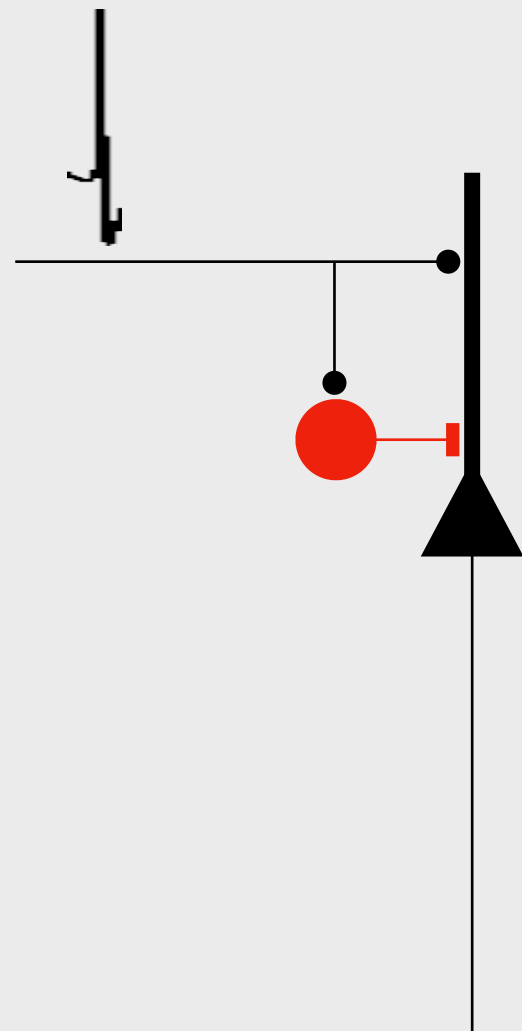
feedback



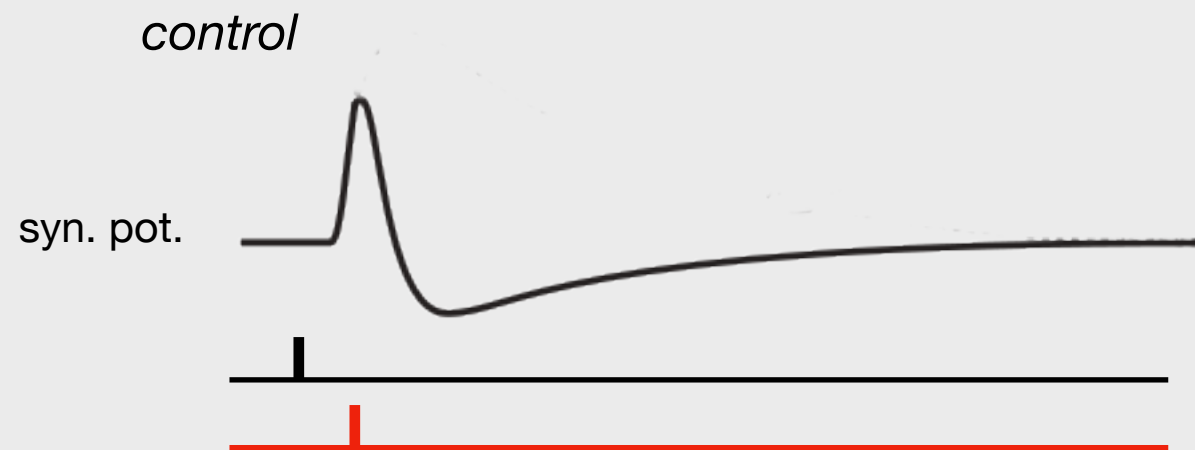
lateral

dendritic conduction & integration

basic inhibitory circuits

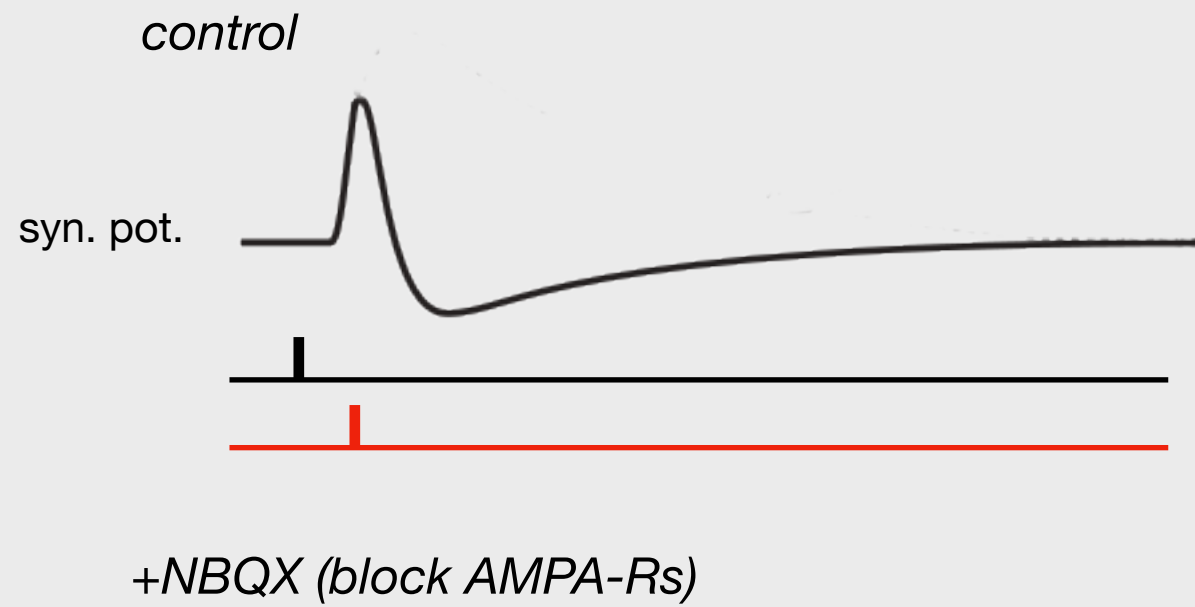
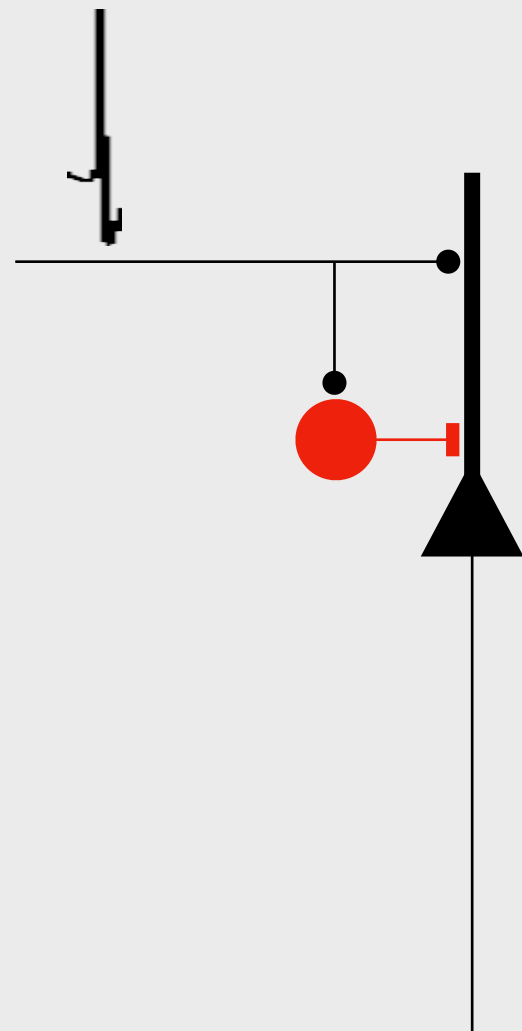


feedforward



dendritic conduction & integration

basic inhibitory circuits

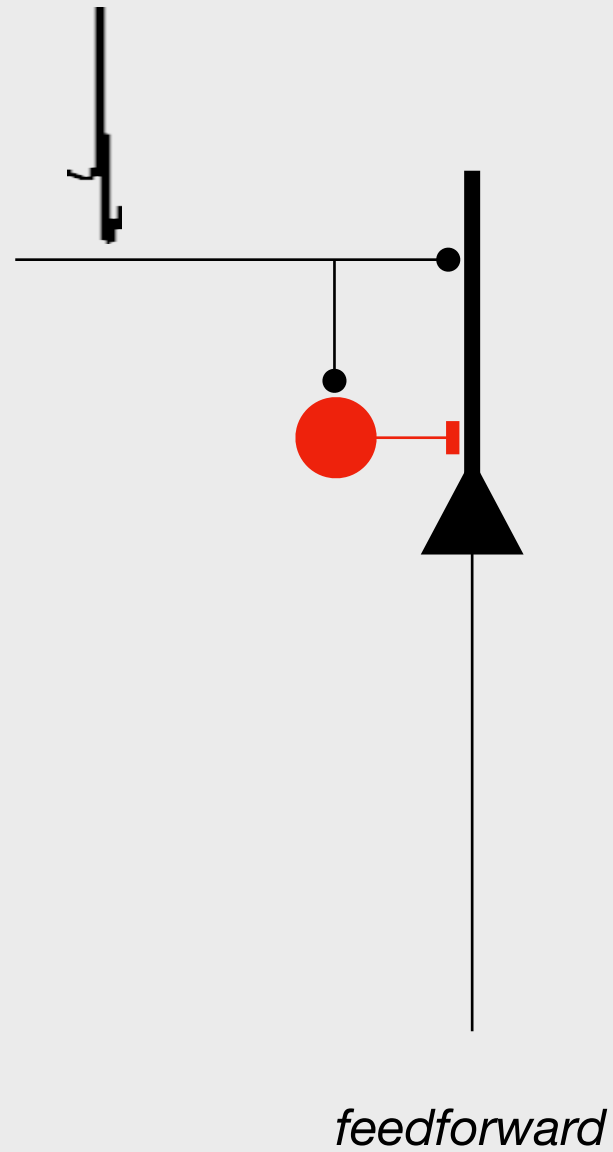


?

feedforward

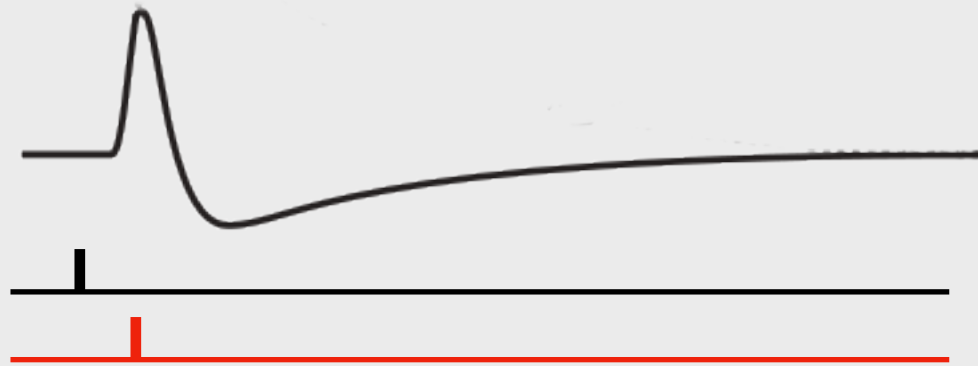
dendritic conduction & integration

basic inhibitory circuits



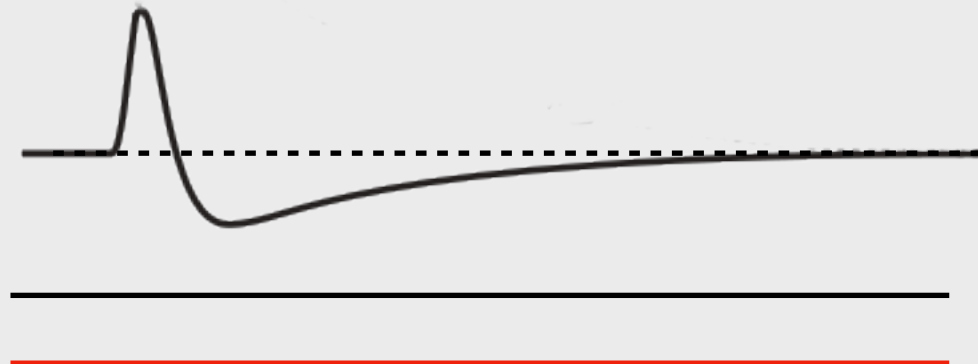
control

syn. pot.



+NBQX

syn. pot.

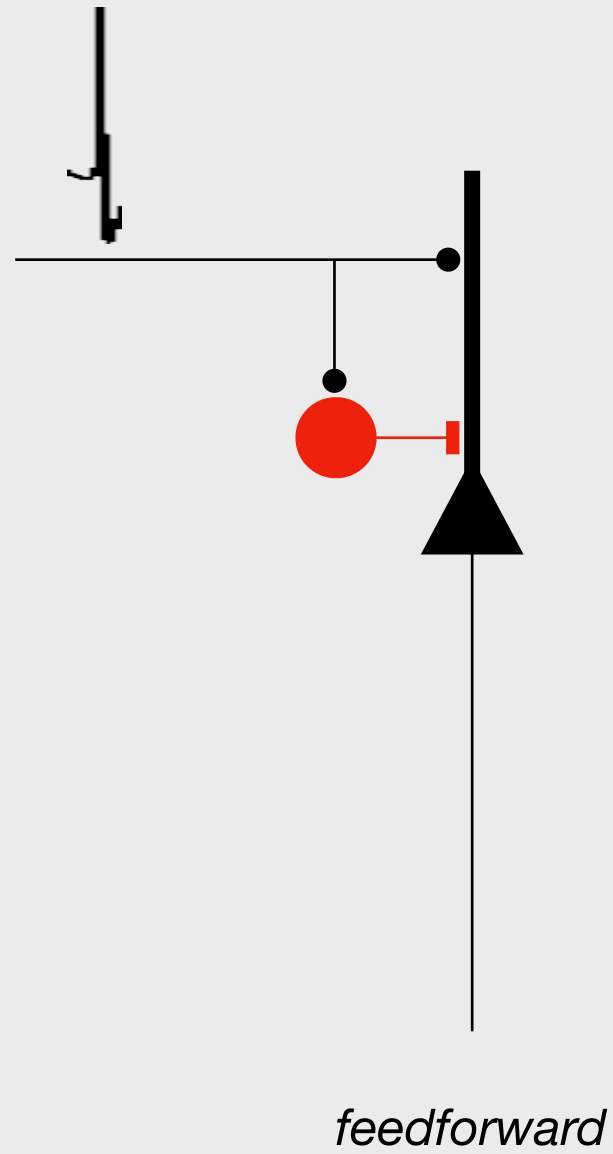


+gabazine (block $GABA_A$ -Rs)

?

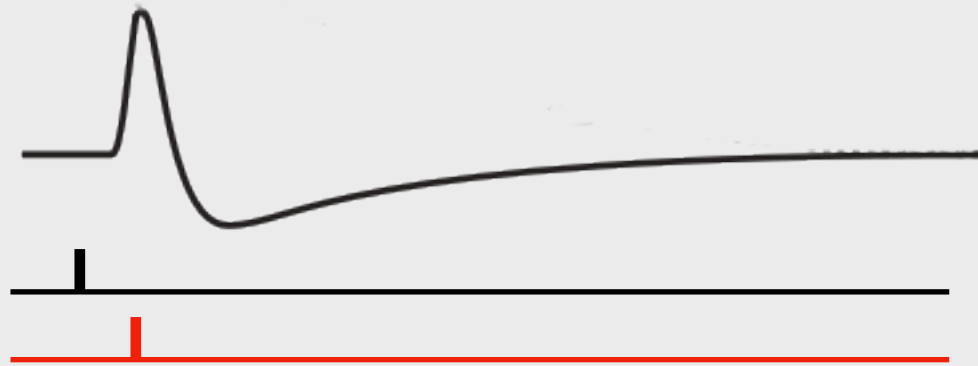
dendritic conduction & integration

basic inhibitory circuits



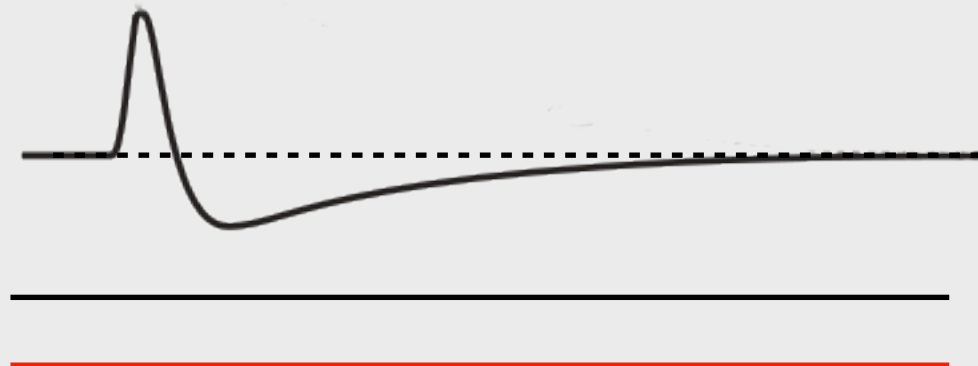
control

syn. pot.



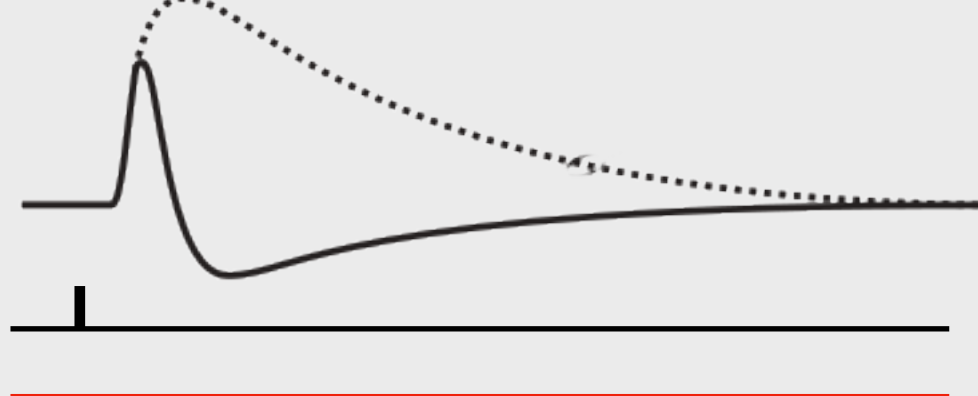
+NBQX

syn. pot.



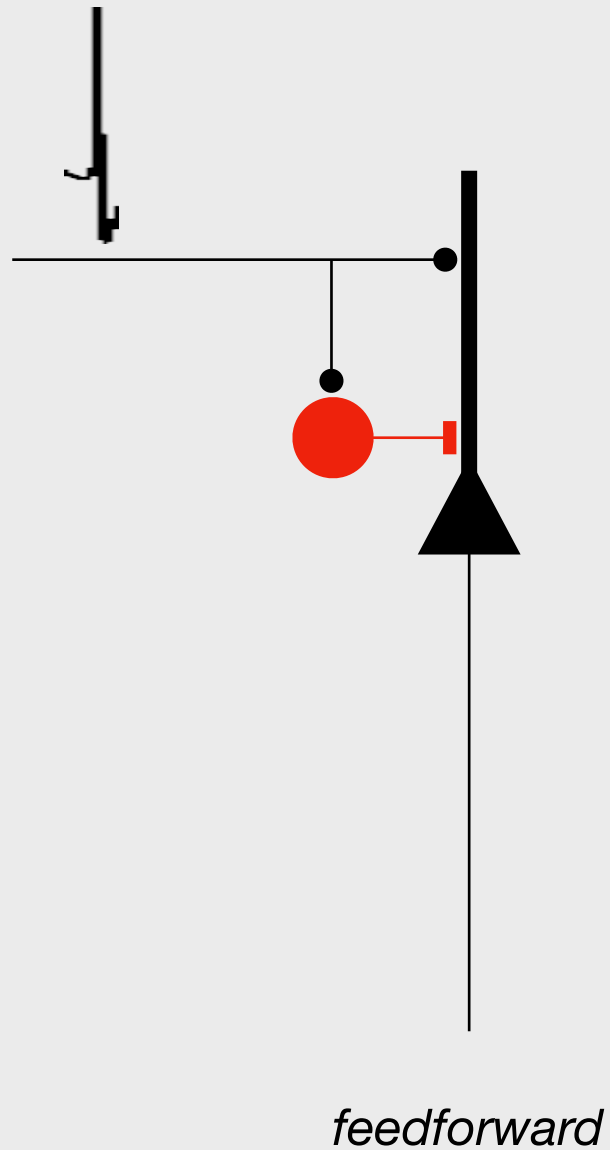
+gabazine

syn. pot.



dendritic conduction & integration

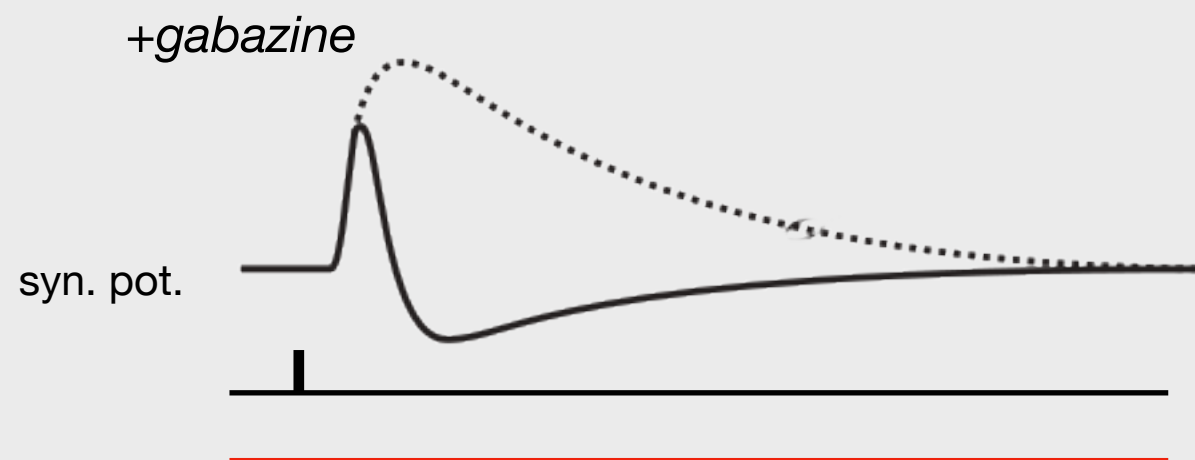
basic inhibitory circuits



feedforward inhibition reduces the integration time window

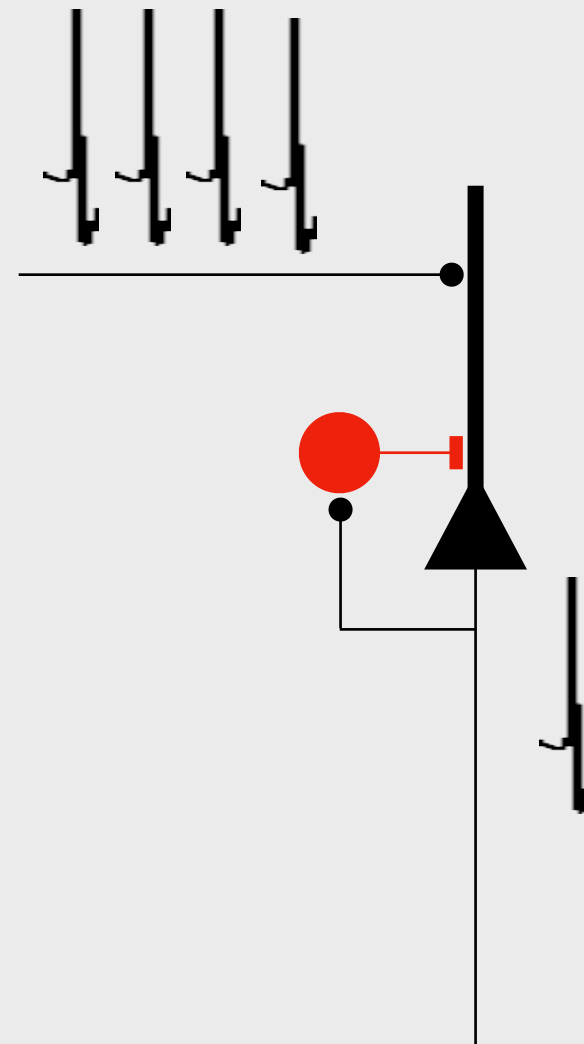


requirement of synchronous inputs to activate the post-synaptic neuron



dendritic conduction & integration

basic inhibitory circuits



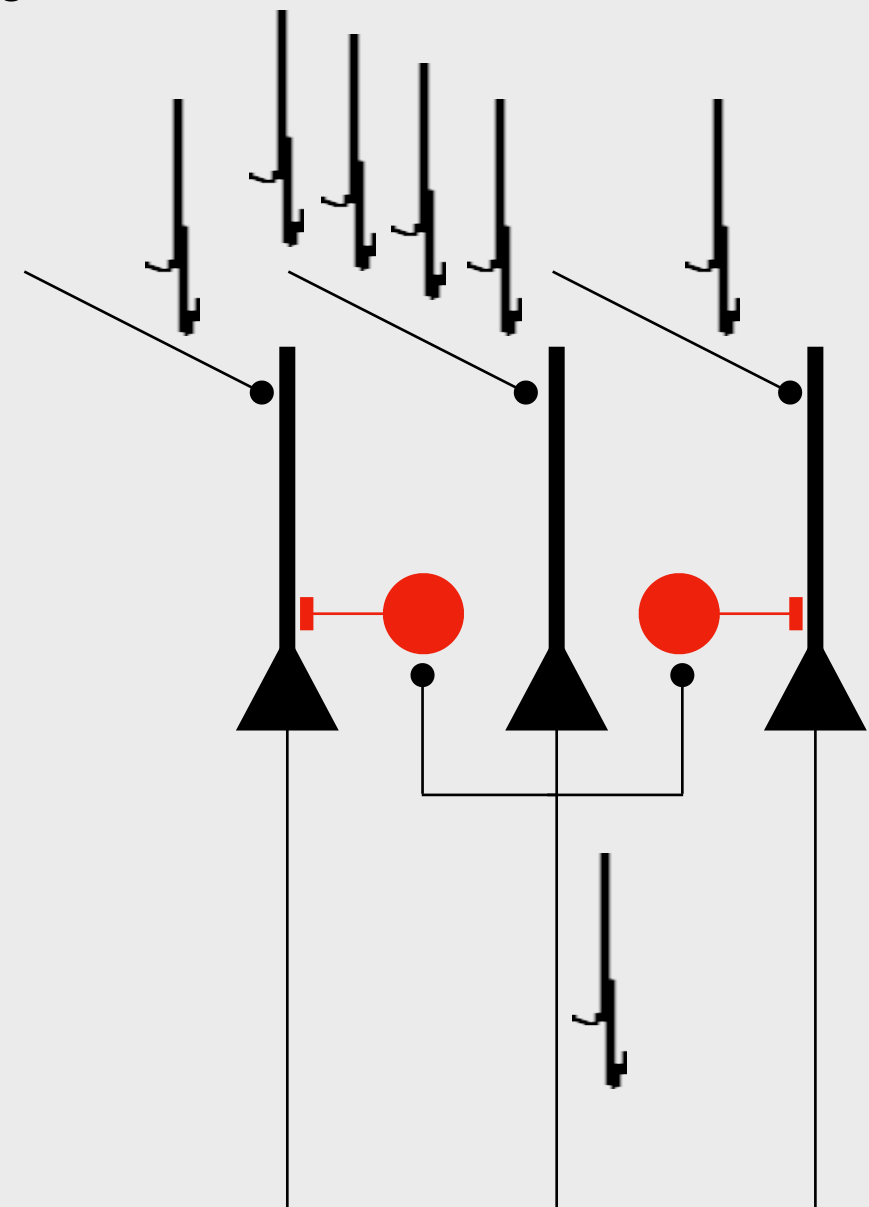
dampen the output of the post-synaptic neuron

feedback

dendritic conduction & integration

basic inhibitory circuits

increase the « contrast » of the output



lateral